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and of which one electrode is electrically connected to a non-exposed surface of the second electrode.

12 Claims, 29 Drawing Sheets

(51) Int. Cl.

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H04N 25/77 (2023.01)
H04N 25/79 (2023.01)
H10D 1/68 (2025.01)
H10F 39/00 (2025.01)

(52) U.S. Cl.

CPC **H04N 25/59** (2023.01); **H04N 25/77** (2023.01); **H04N 25/79** (2023.01); **H10D 1/692** (2025.01); **H10F 39/802** (2025.01)

(58) Field of Classification Search

CPC H01L 23/522; H01L 27/146; H01L 27/14601; H01L 28/40; H01G 4/008; H01G 4/30; H01G 4/33; H04N 23/54; H04N 25/59; H04N 25/77; H04N 25/79; H04N 25/771; H04N 25/78; H04N 23/55
 USPC 257/291
 See application file for complete search history.

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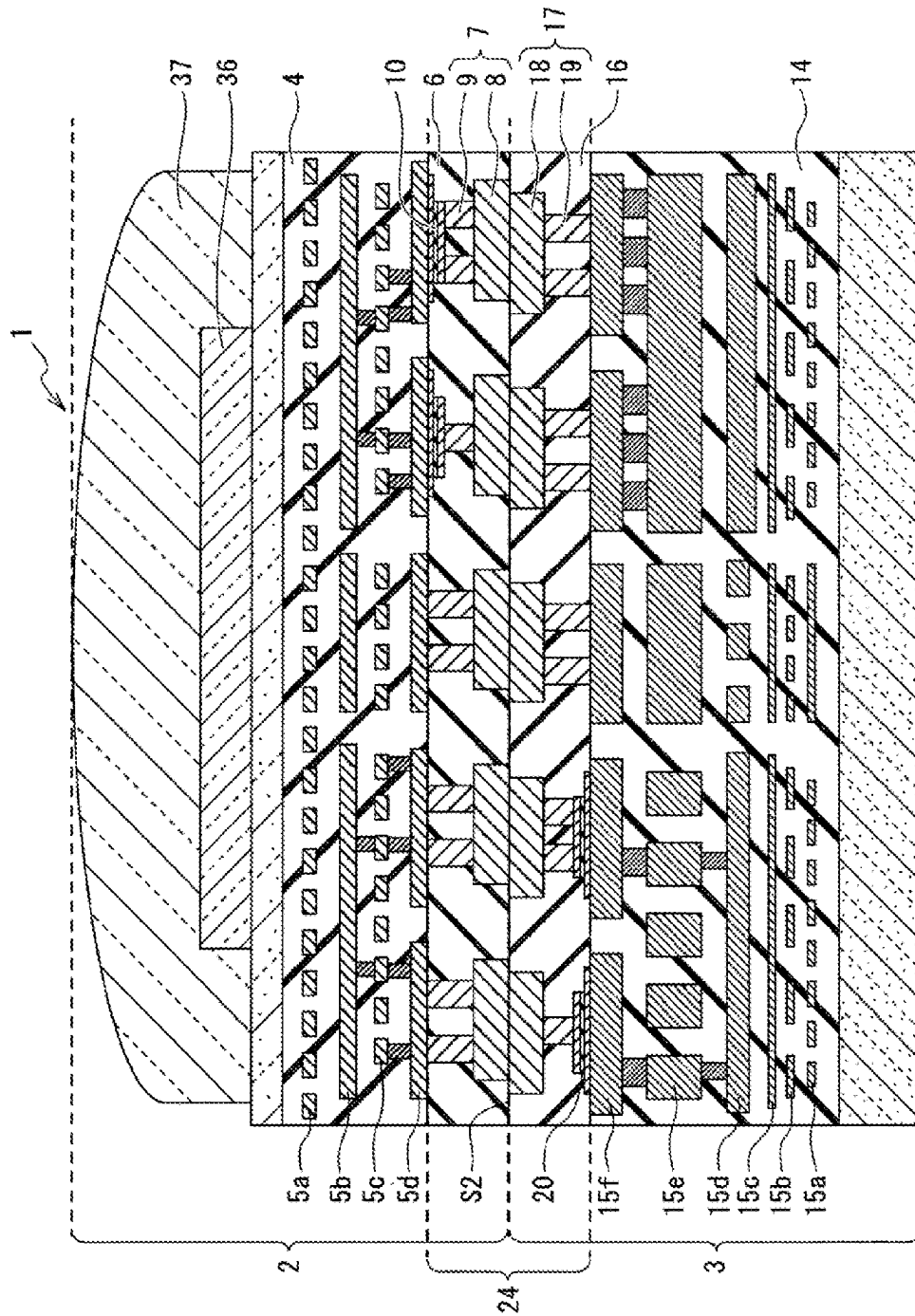


Fig. 1

Fig. 2

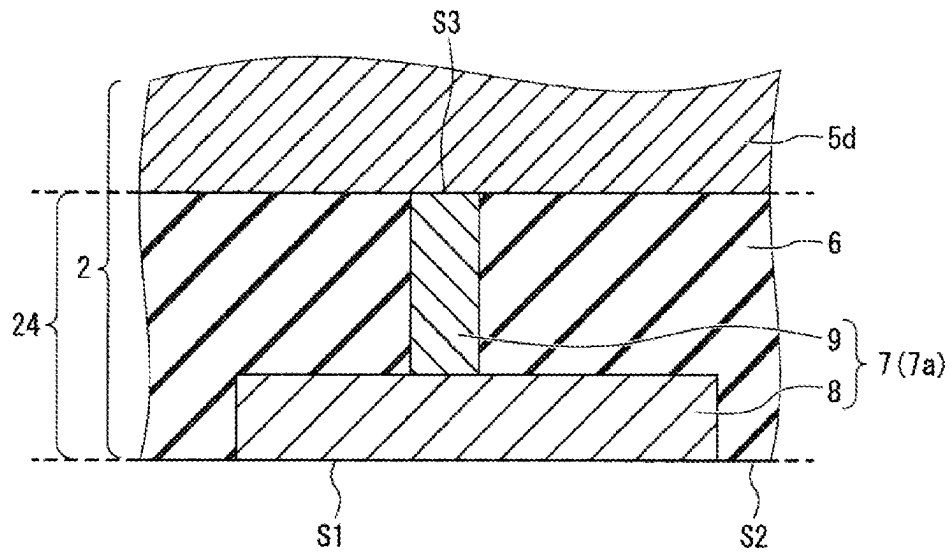


Fig. 3

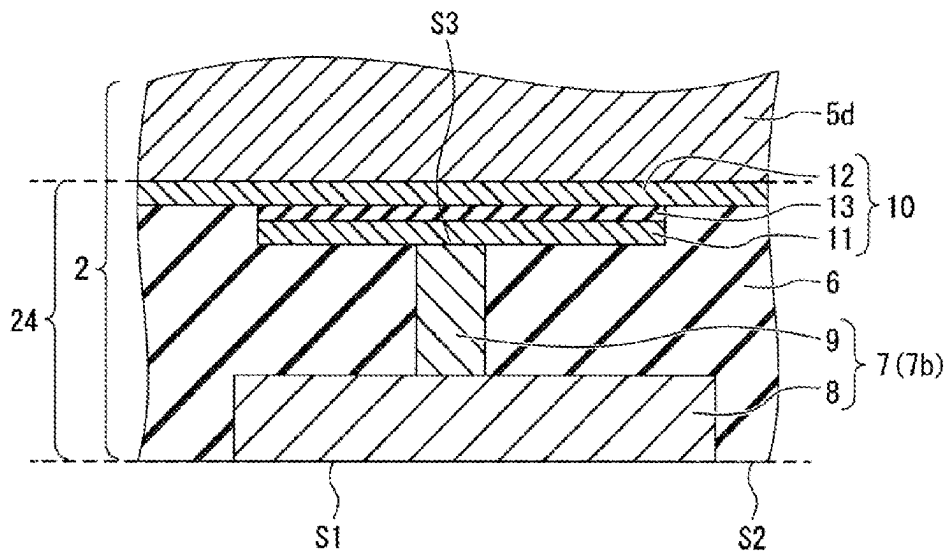


Fig. 4

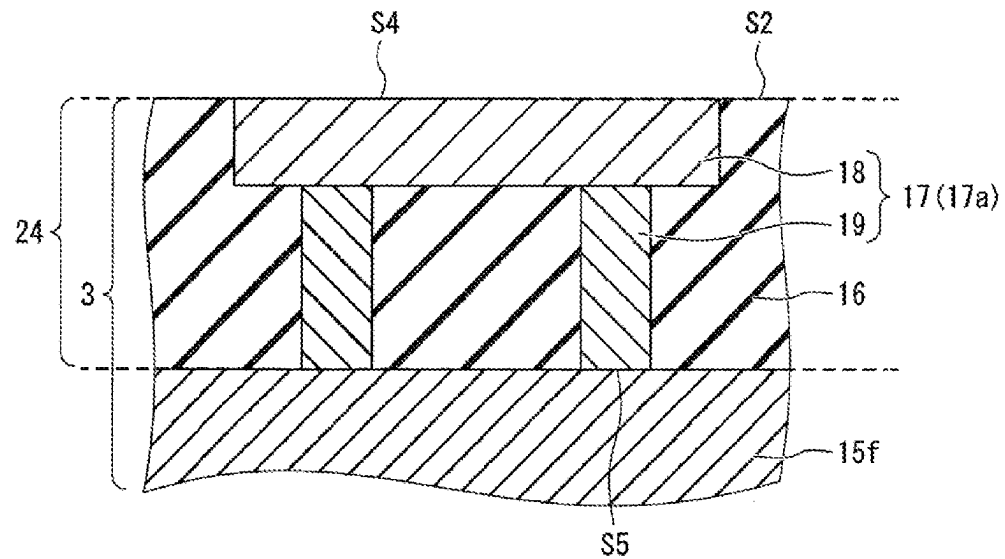


Fig. 5

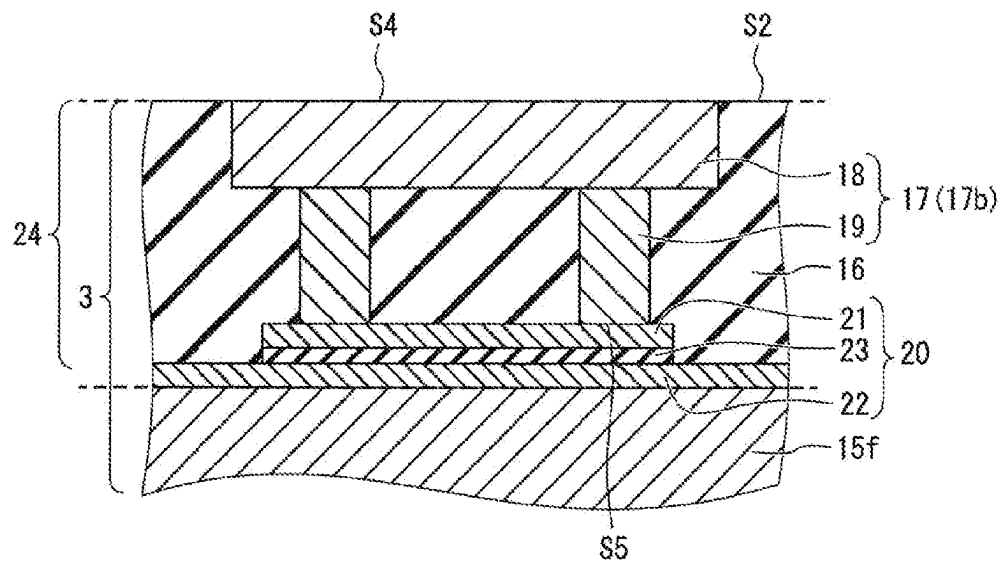


Fig. 6

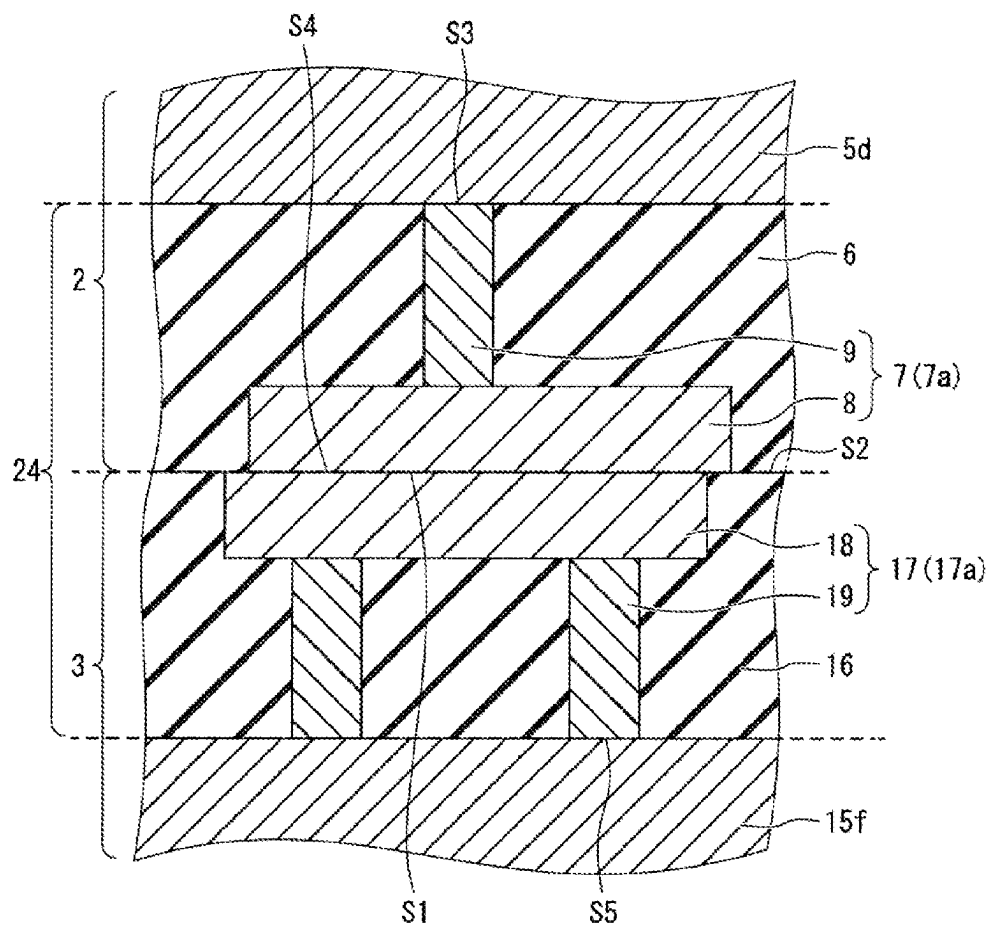


Fig. 7

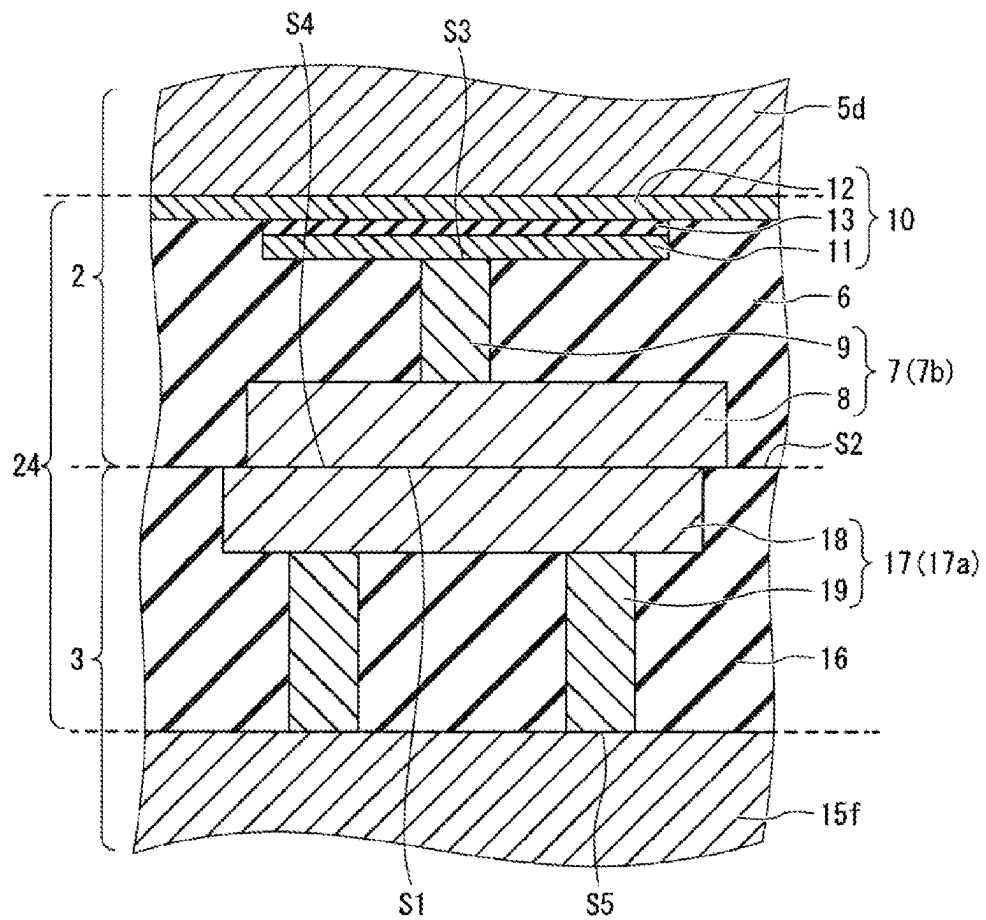


Fig. 8

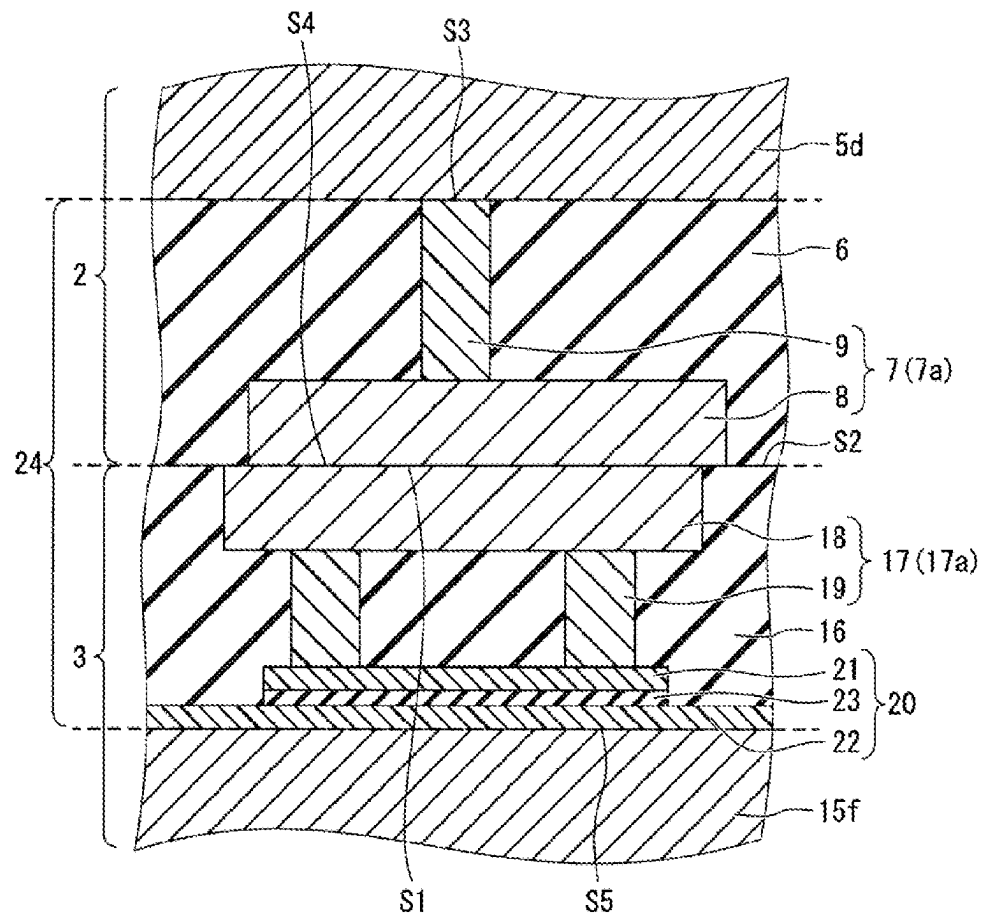


Fig. 9

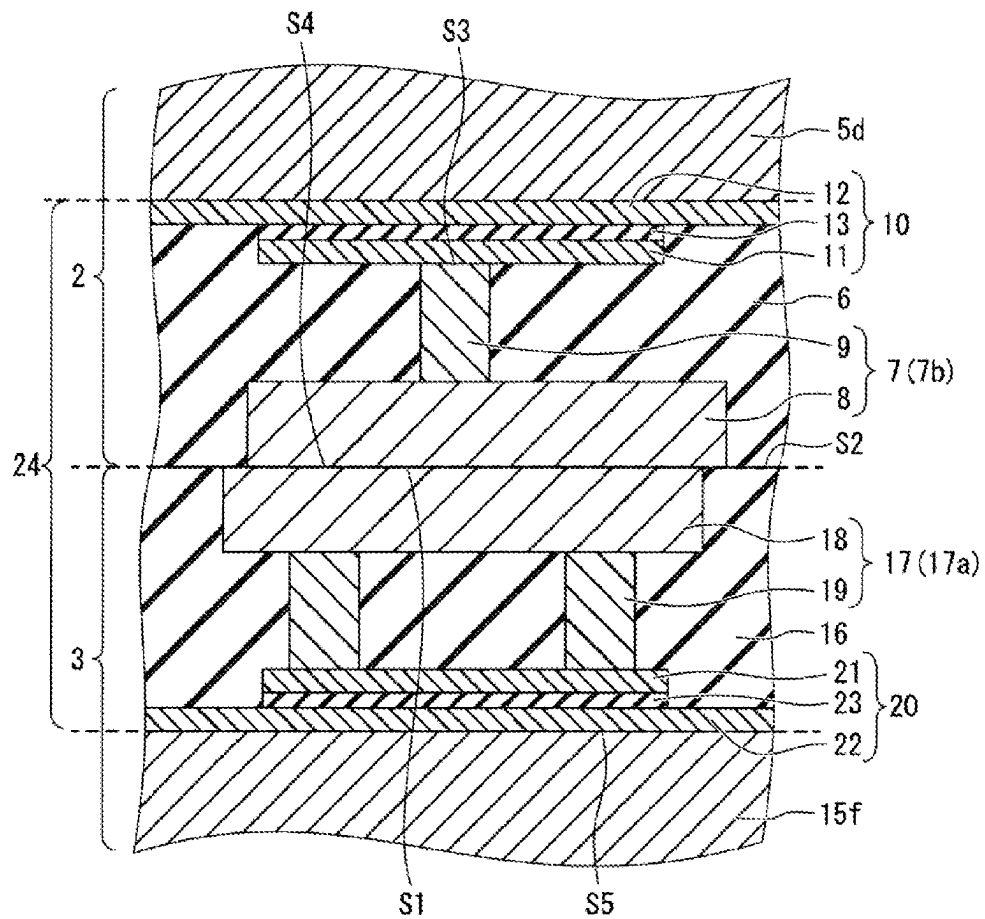


Fig. 10A

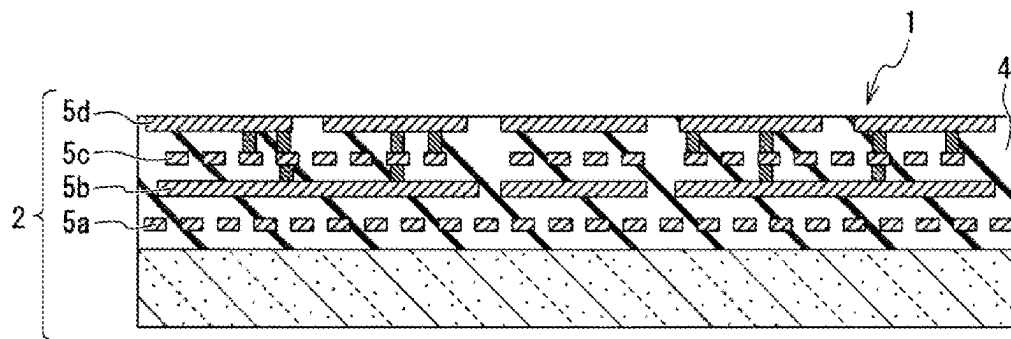
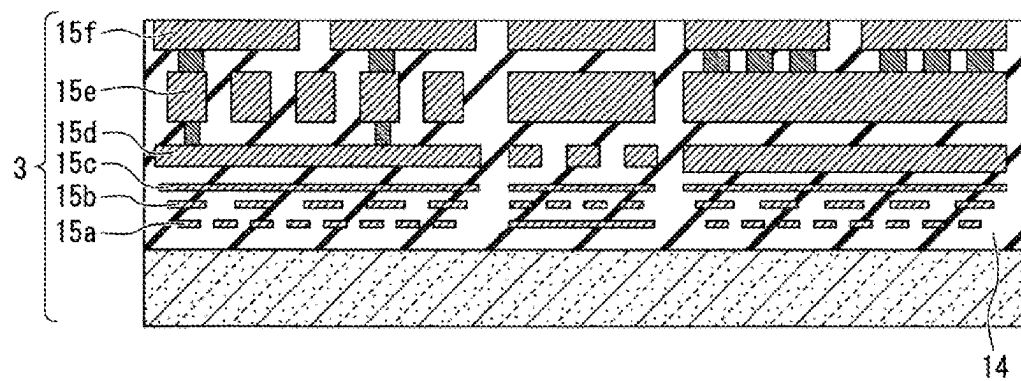


Fig. 10B



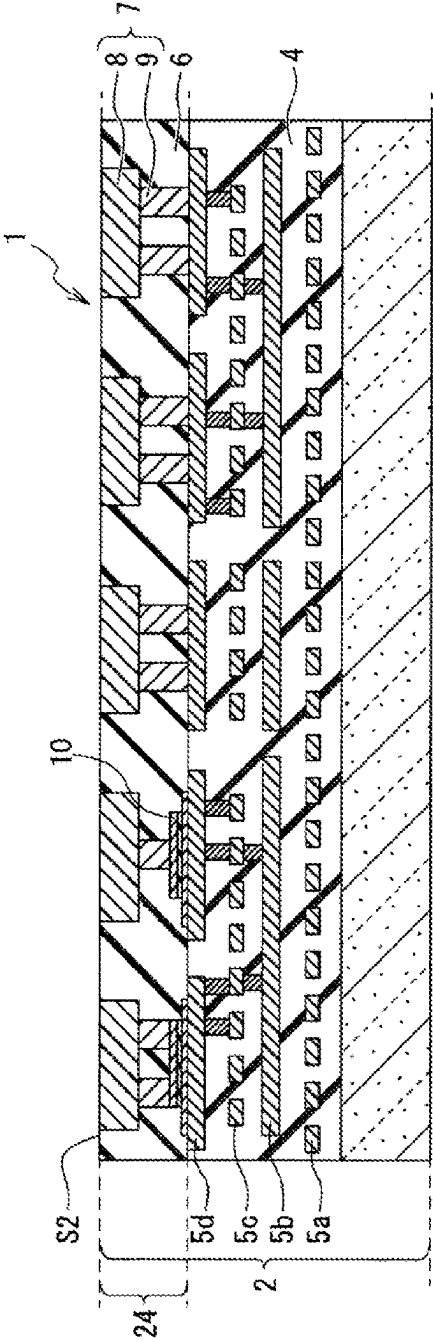


Fig. 11A

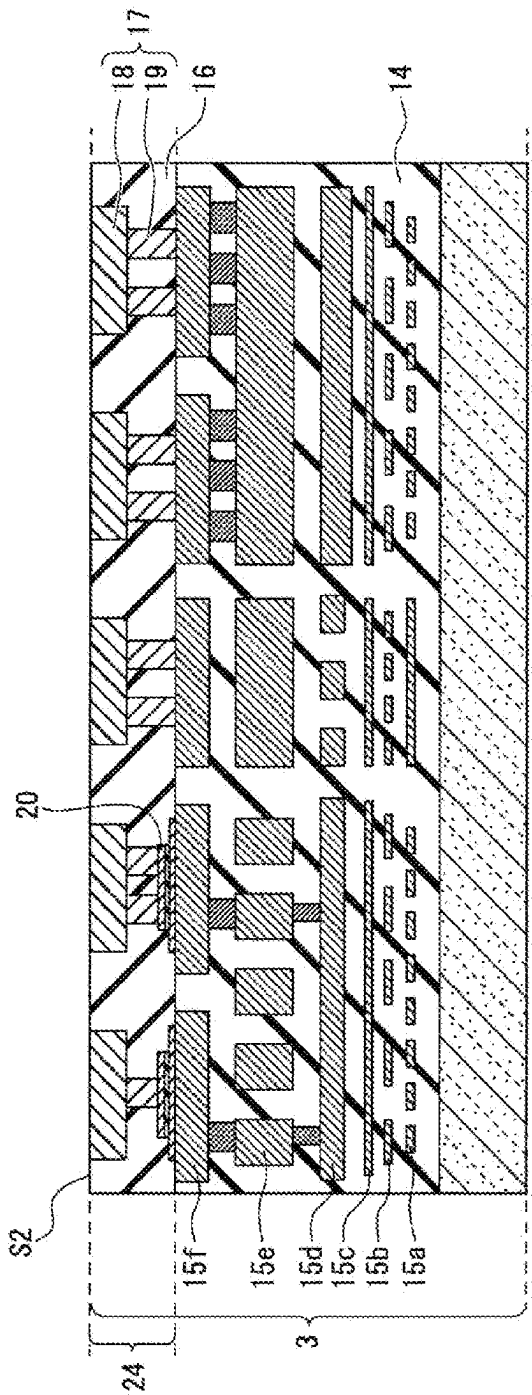
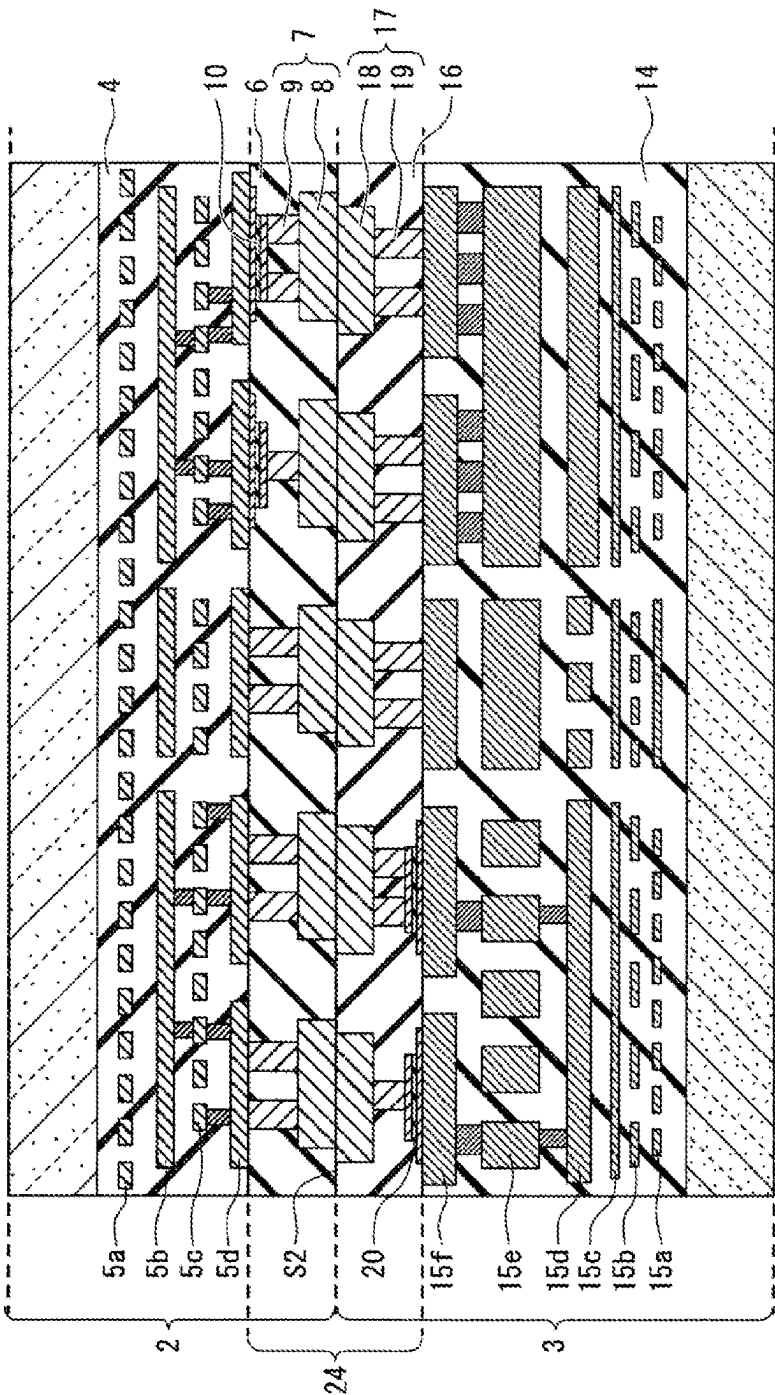


Fig. 11B

Fig. 12



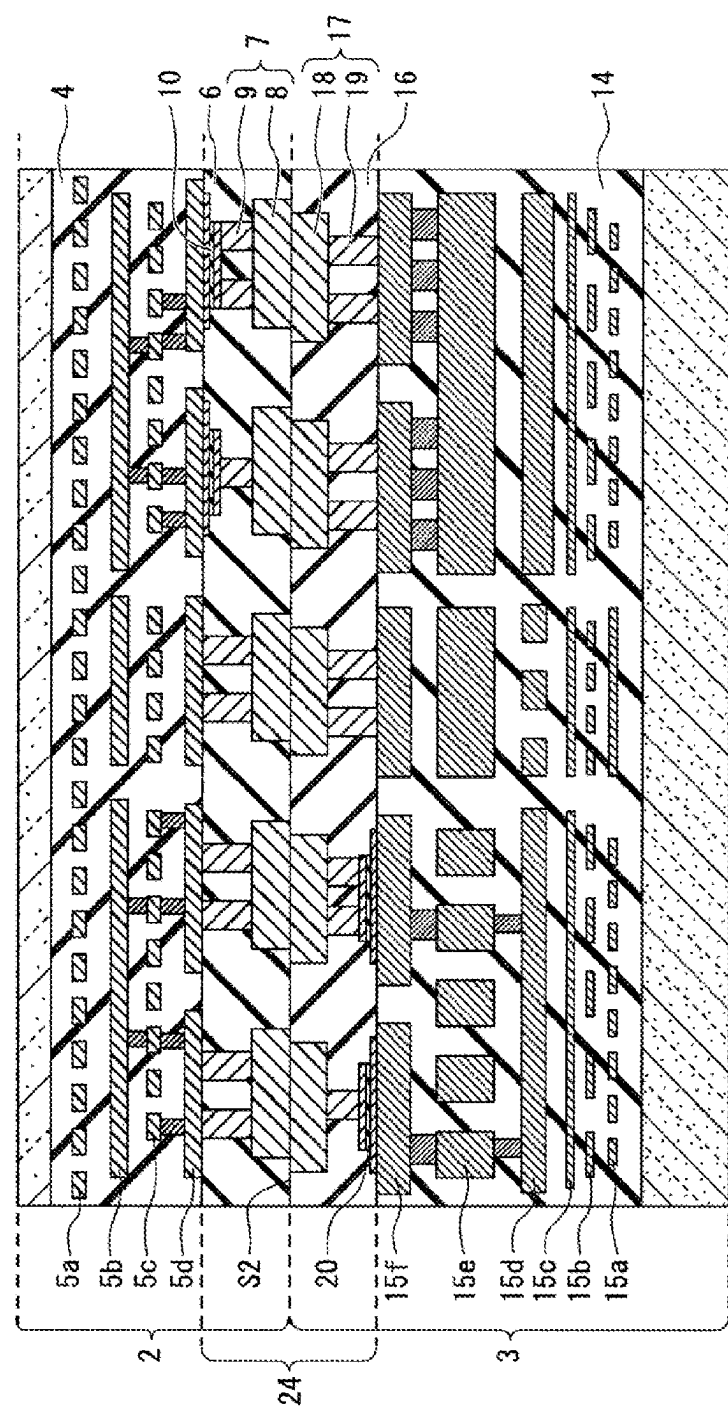


Fig. 13

Fig. 14

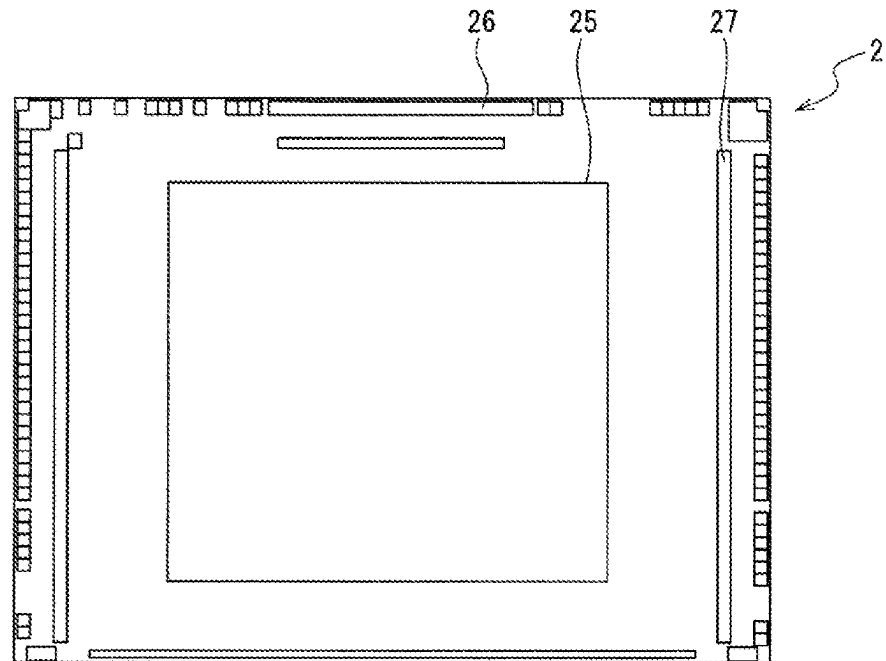


Fig. 15

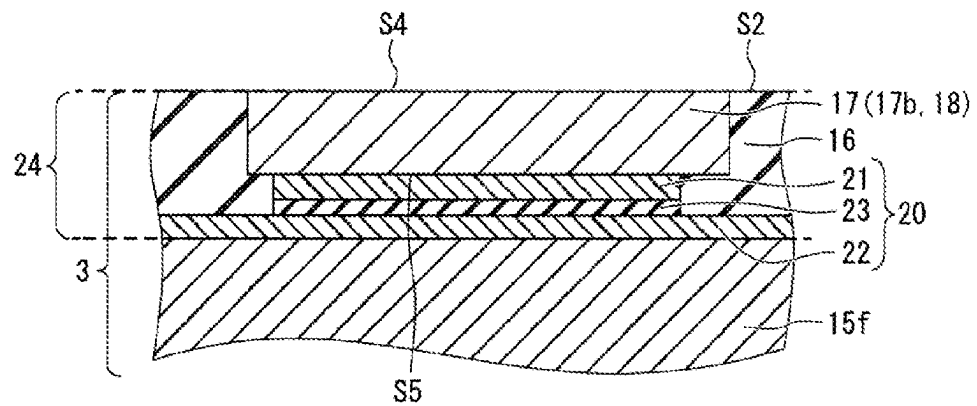


Fig. 16

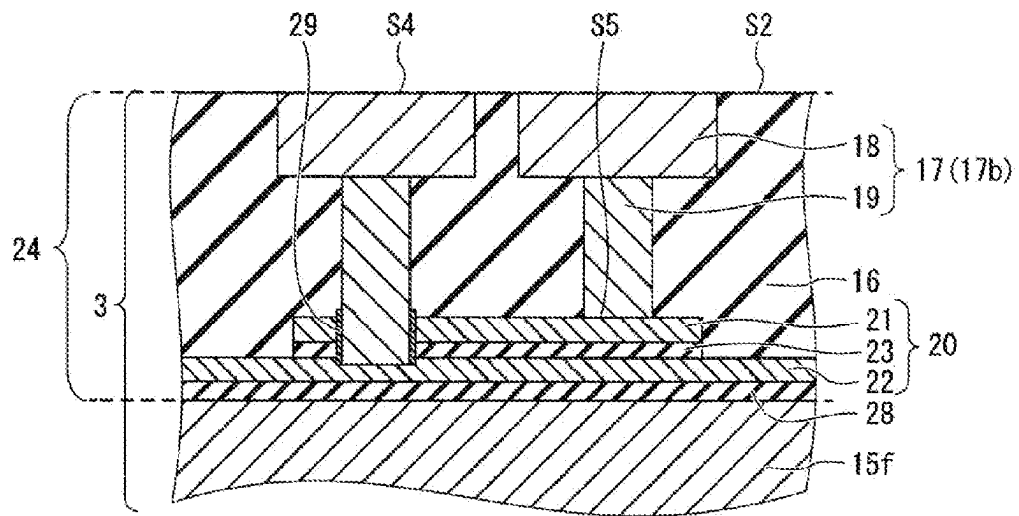


Fig. 17

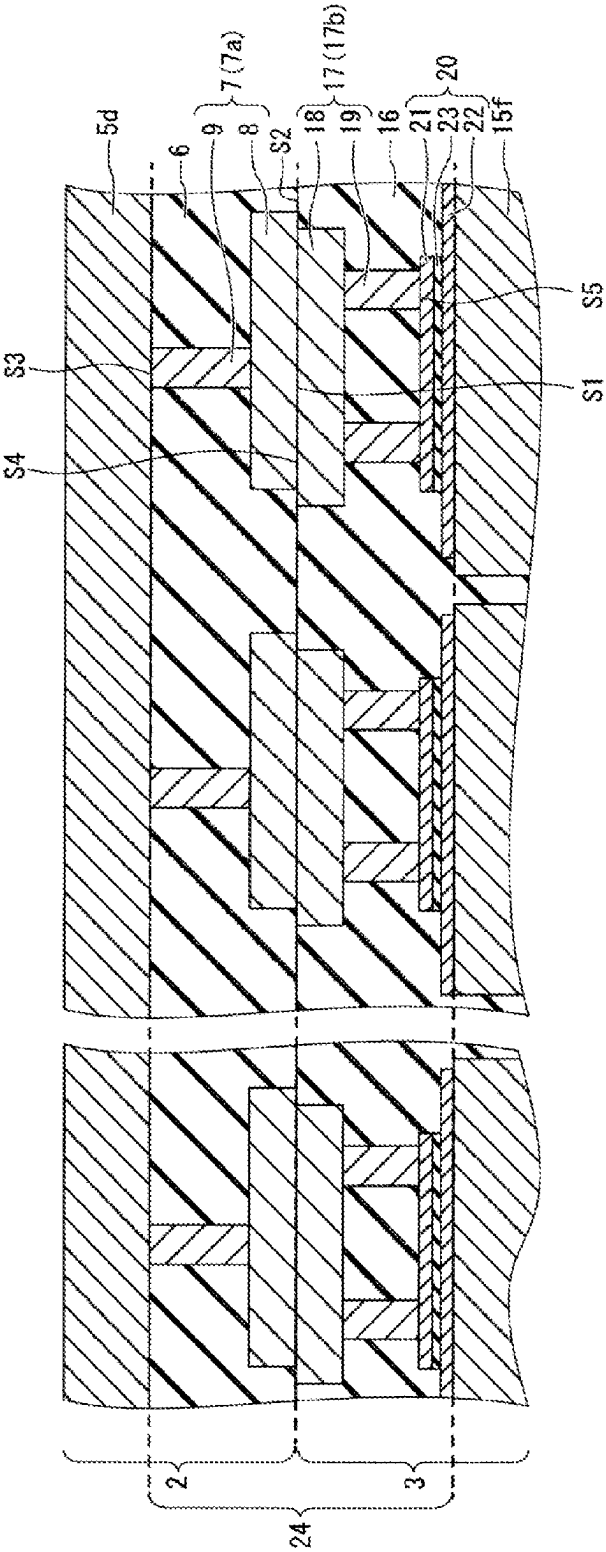


Fig. 18A

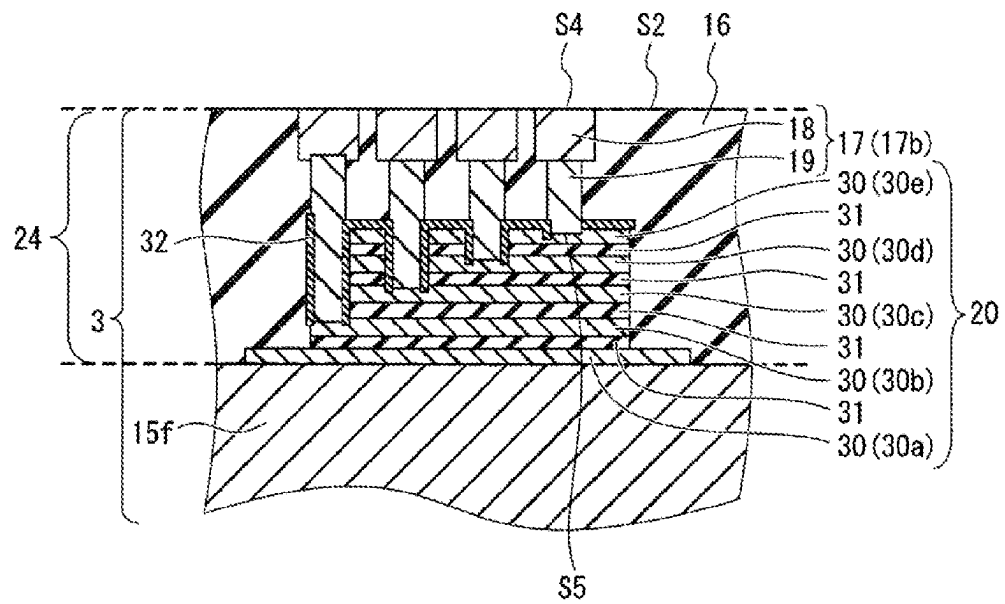


Fig. 18B

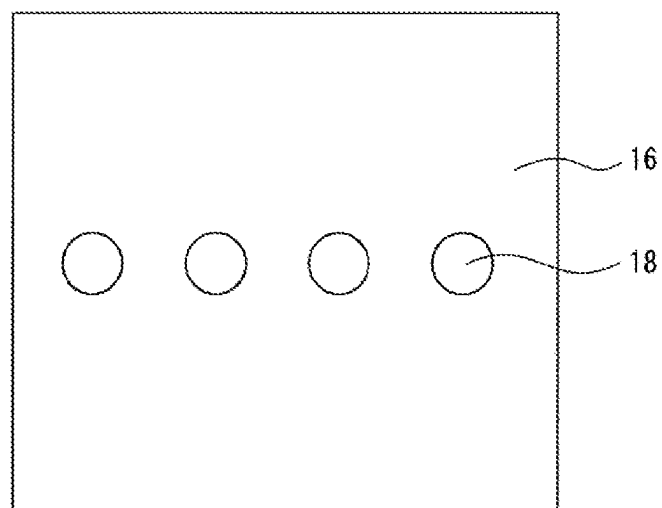


Fig. 19

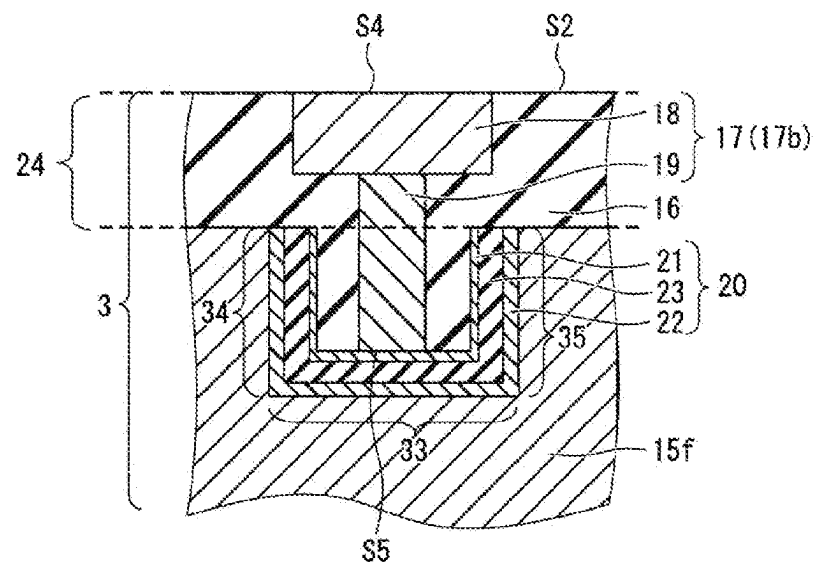


Fig. 20

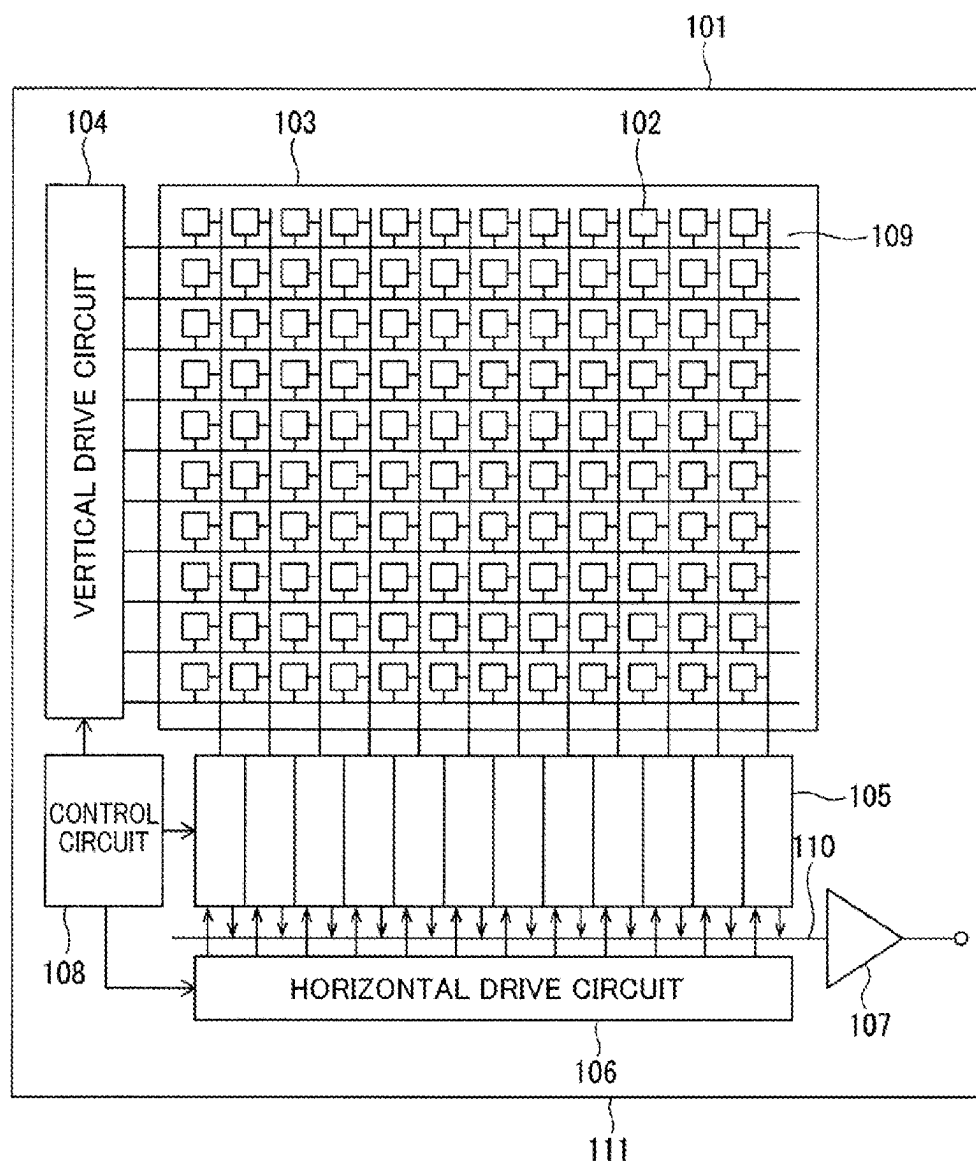


Fig. 21

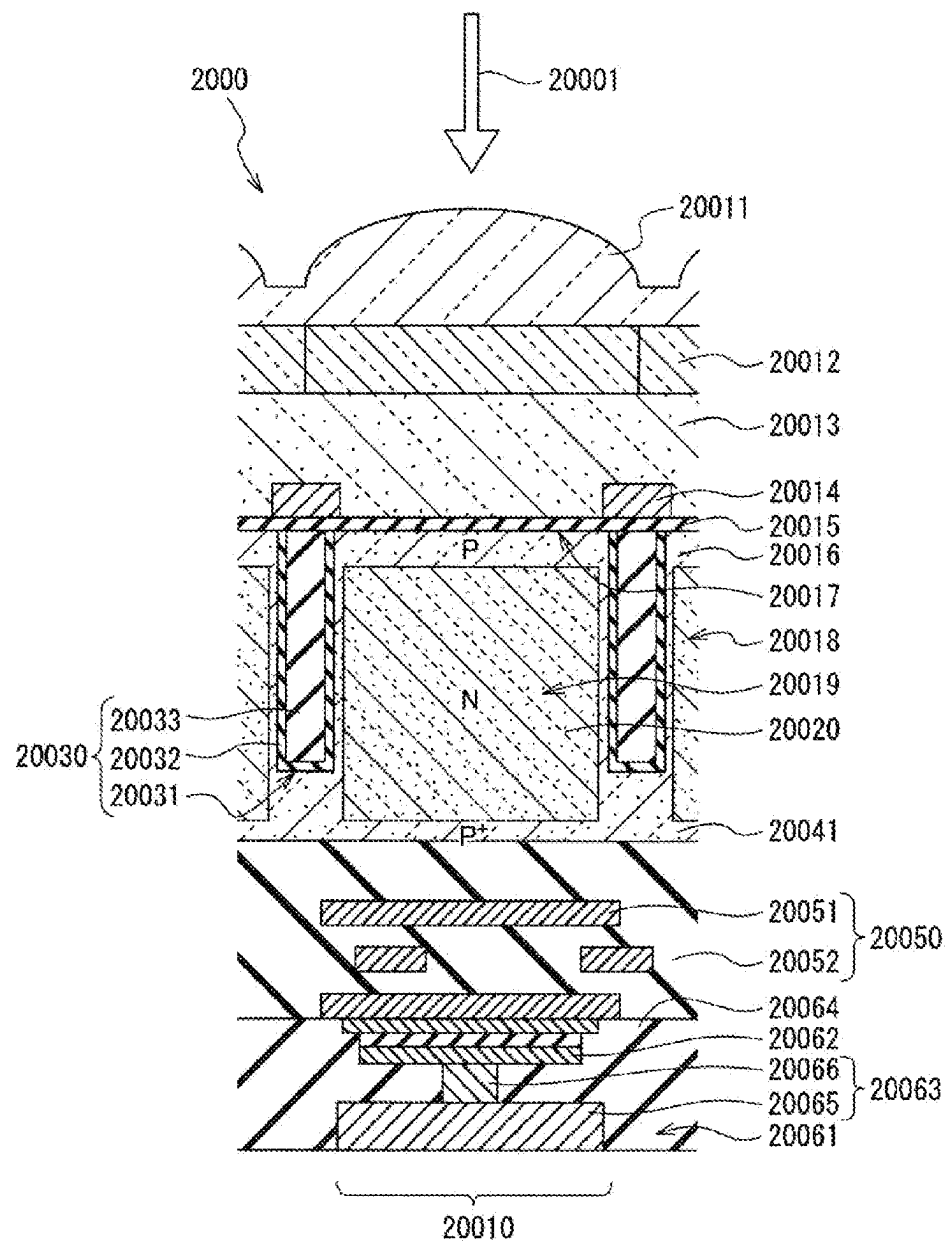


Fig. 22

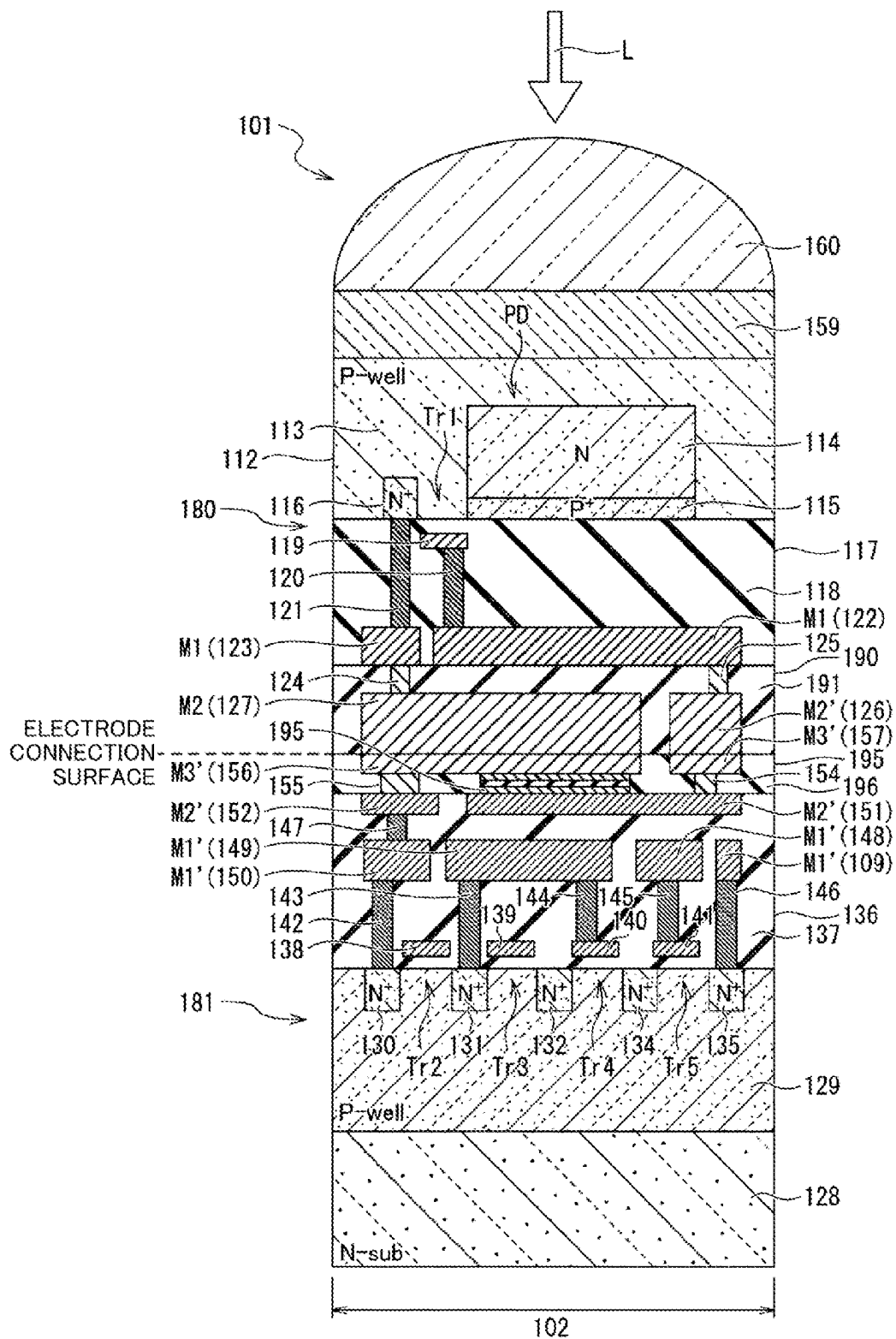


Fig. 23A

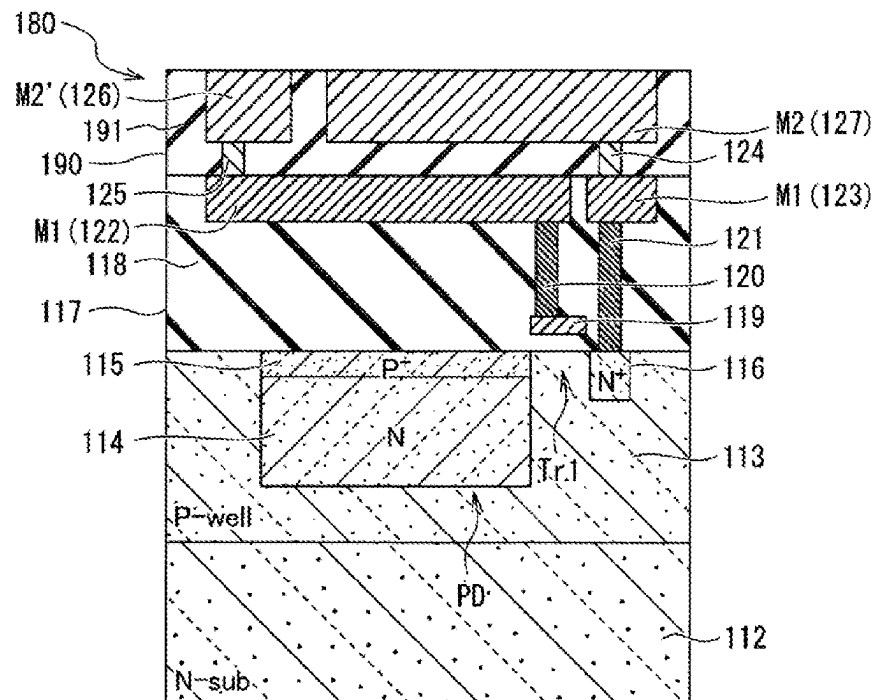


Fig. 23B

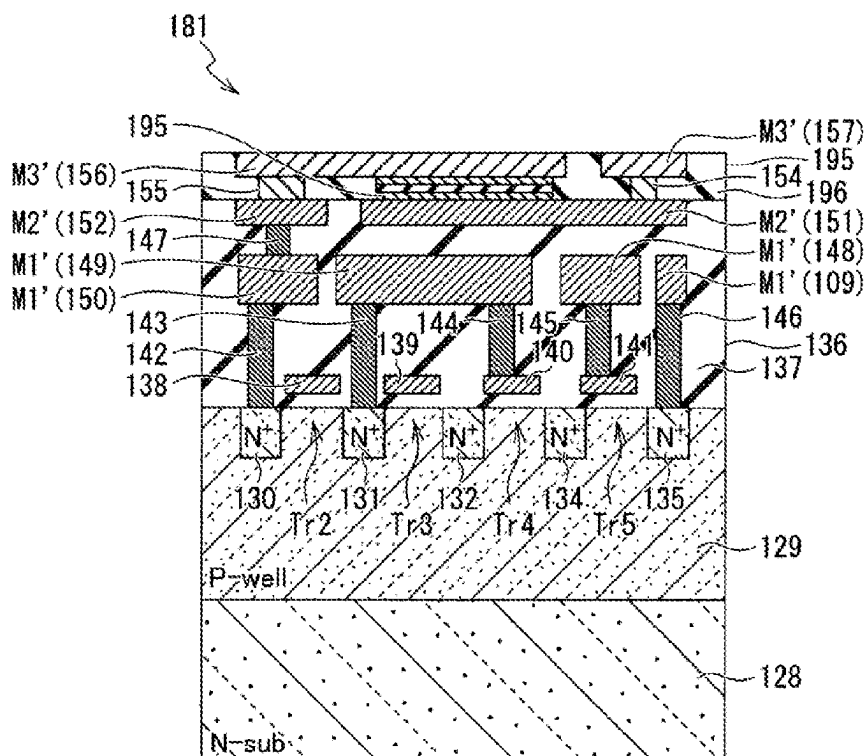


Fig. 24

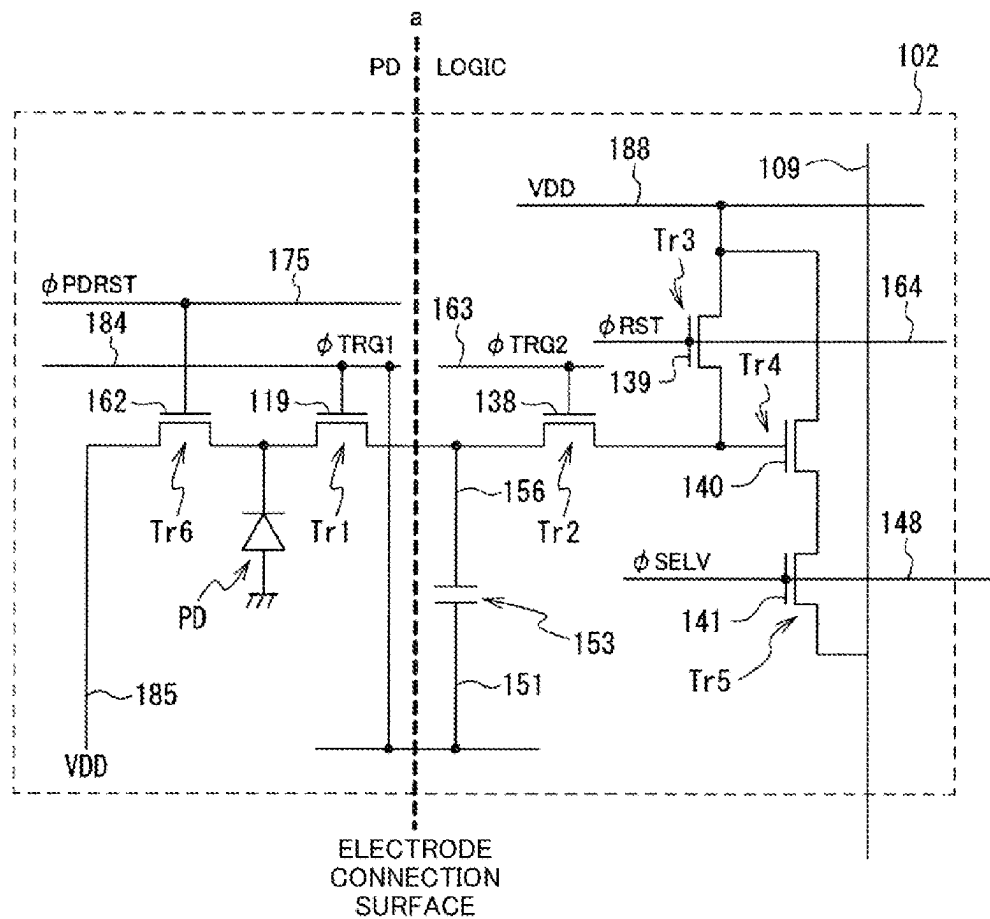


Fig. 25

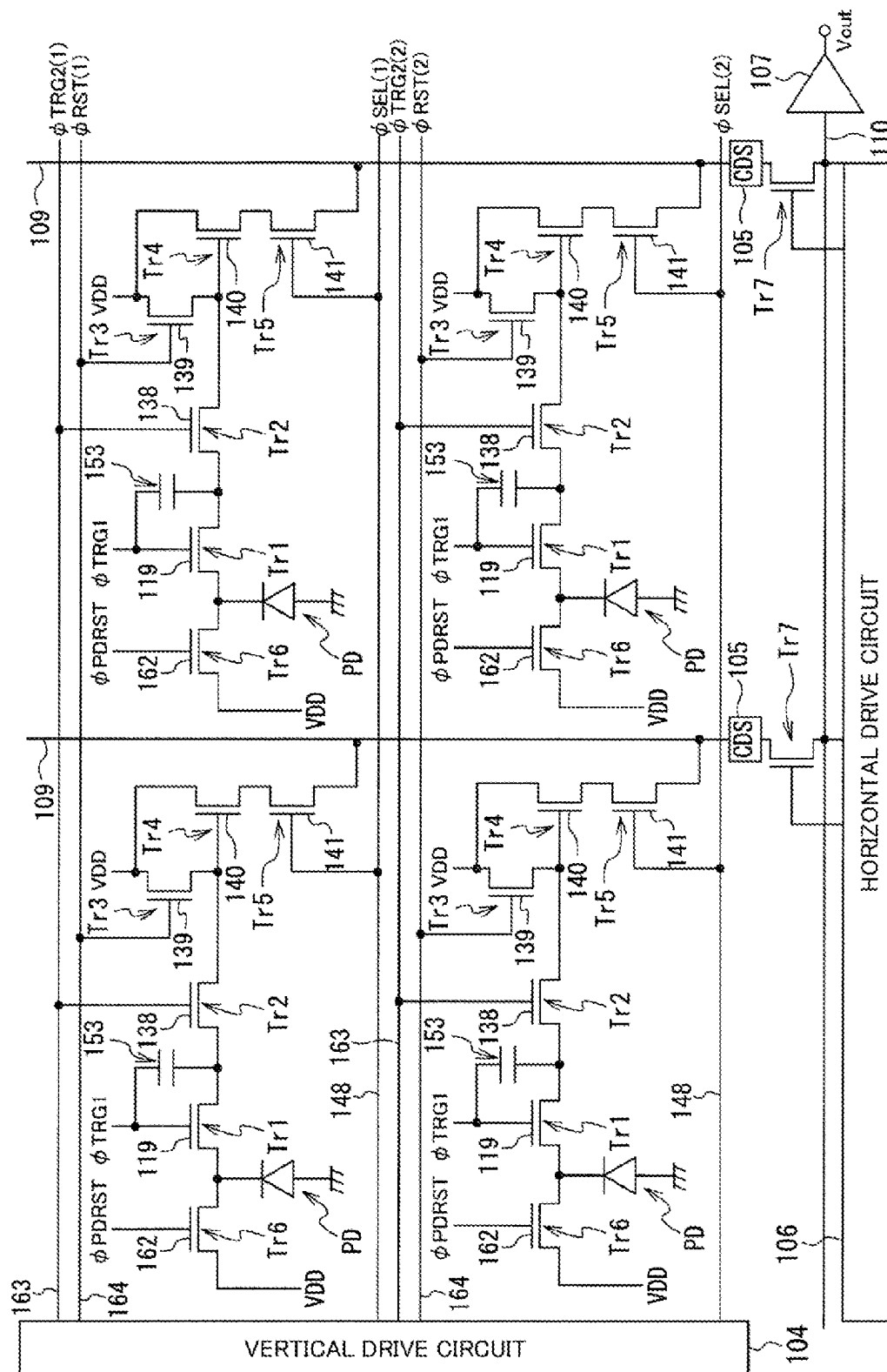


Fig. 26

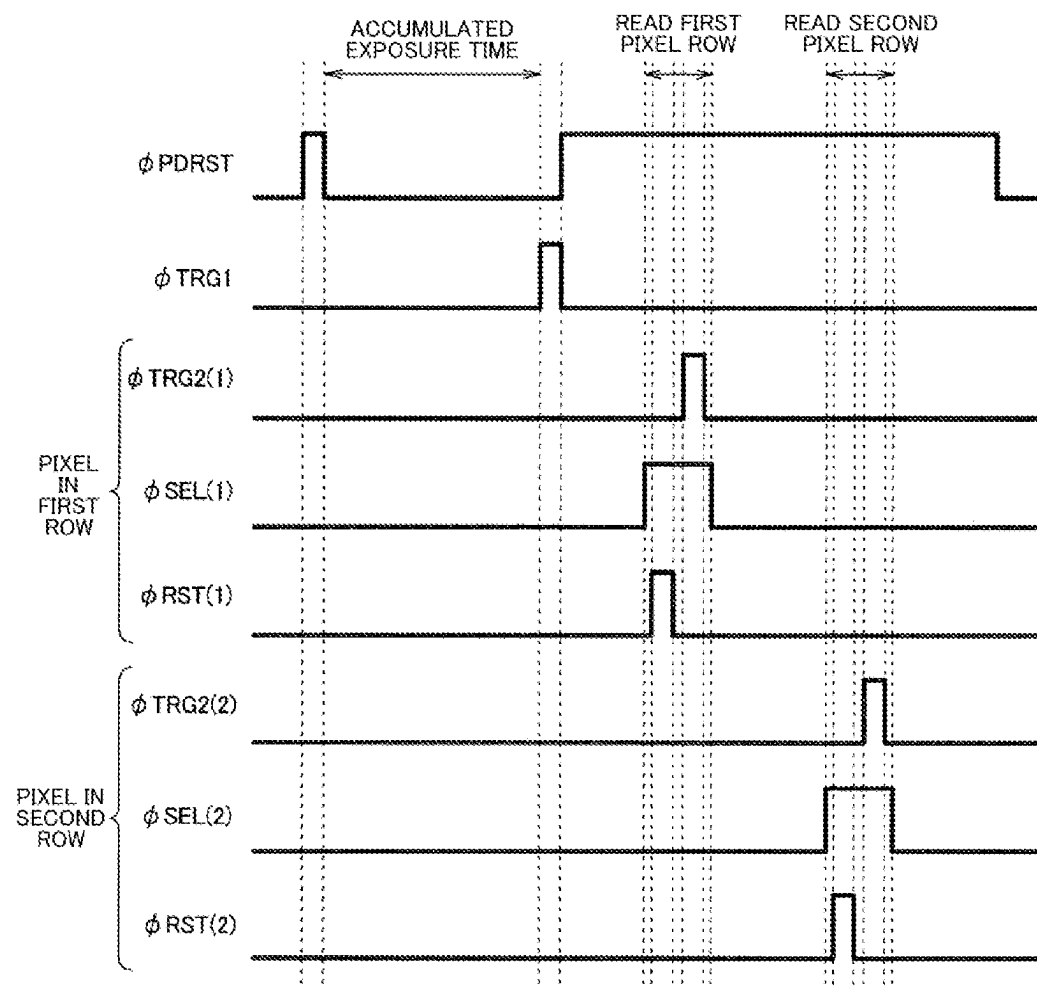
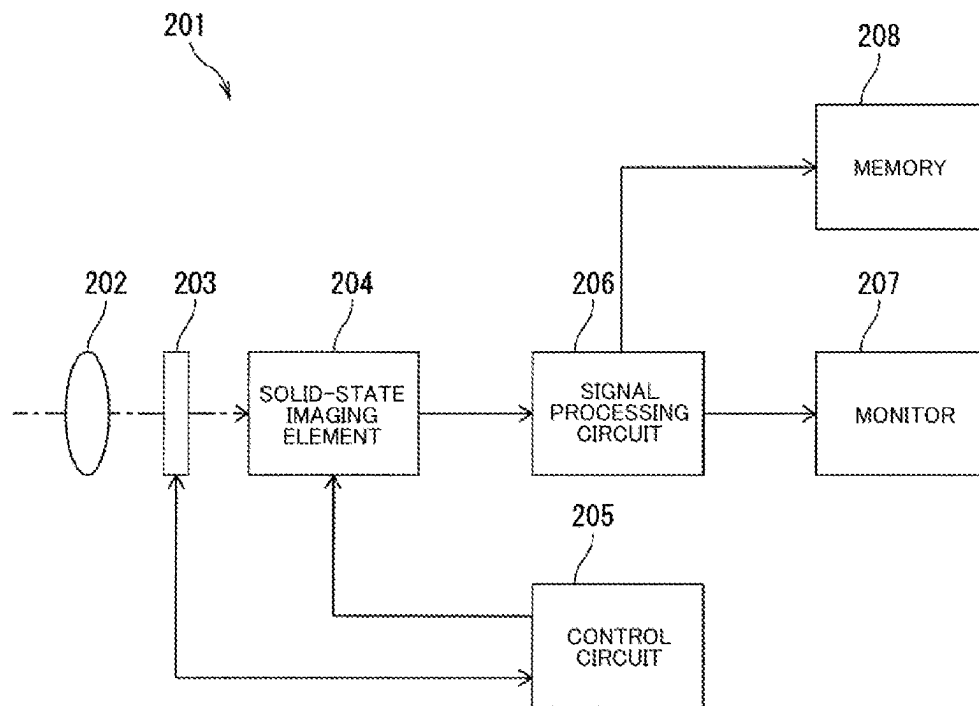


Fig. 27



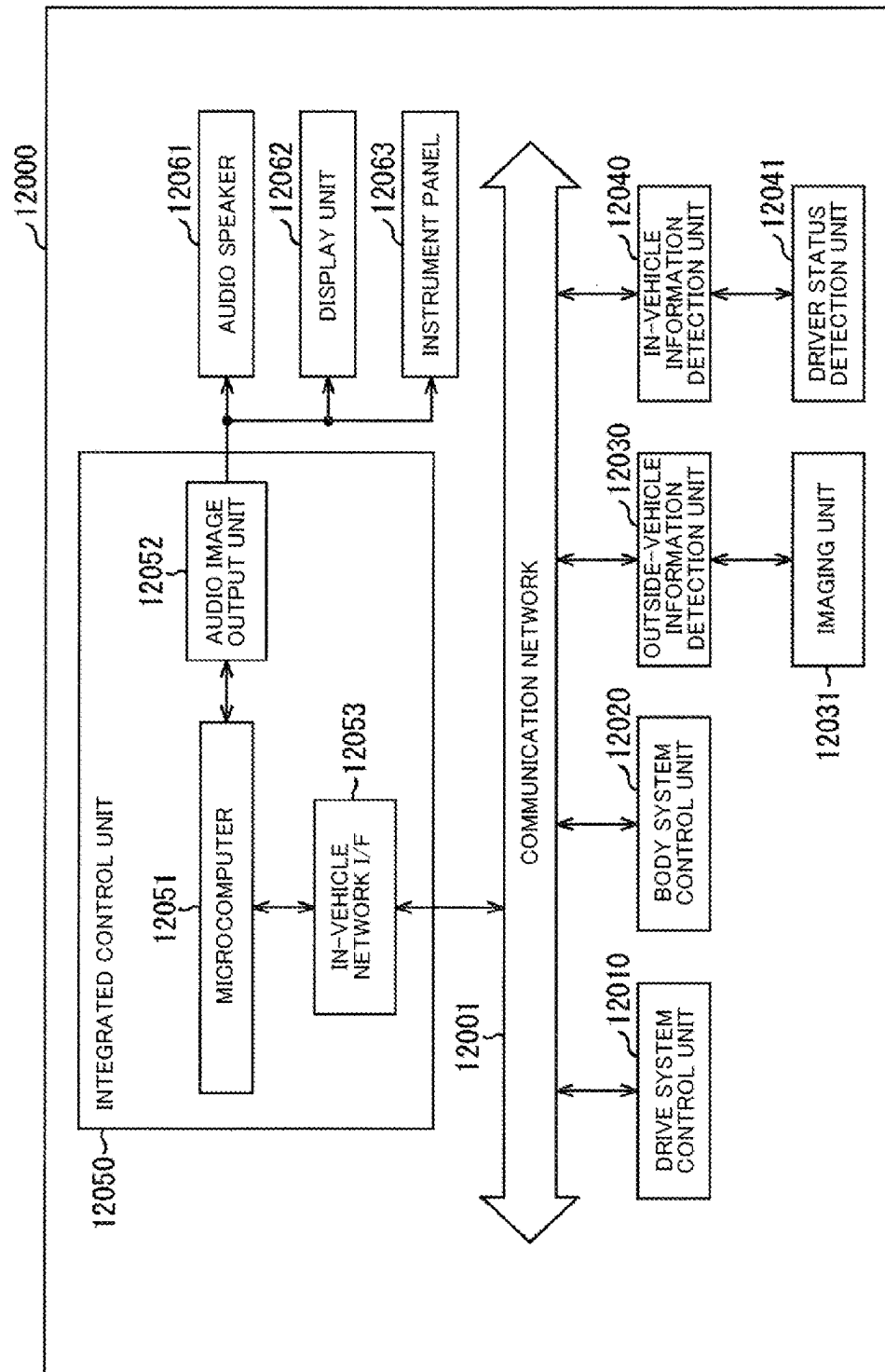
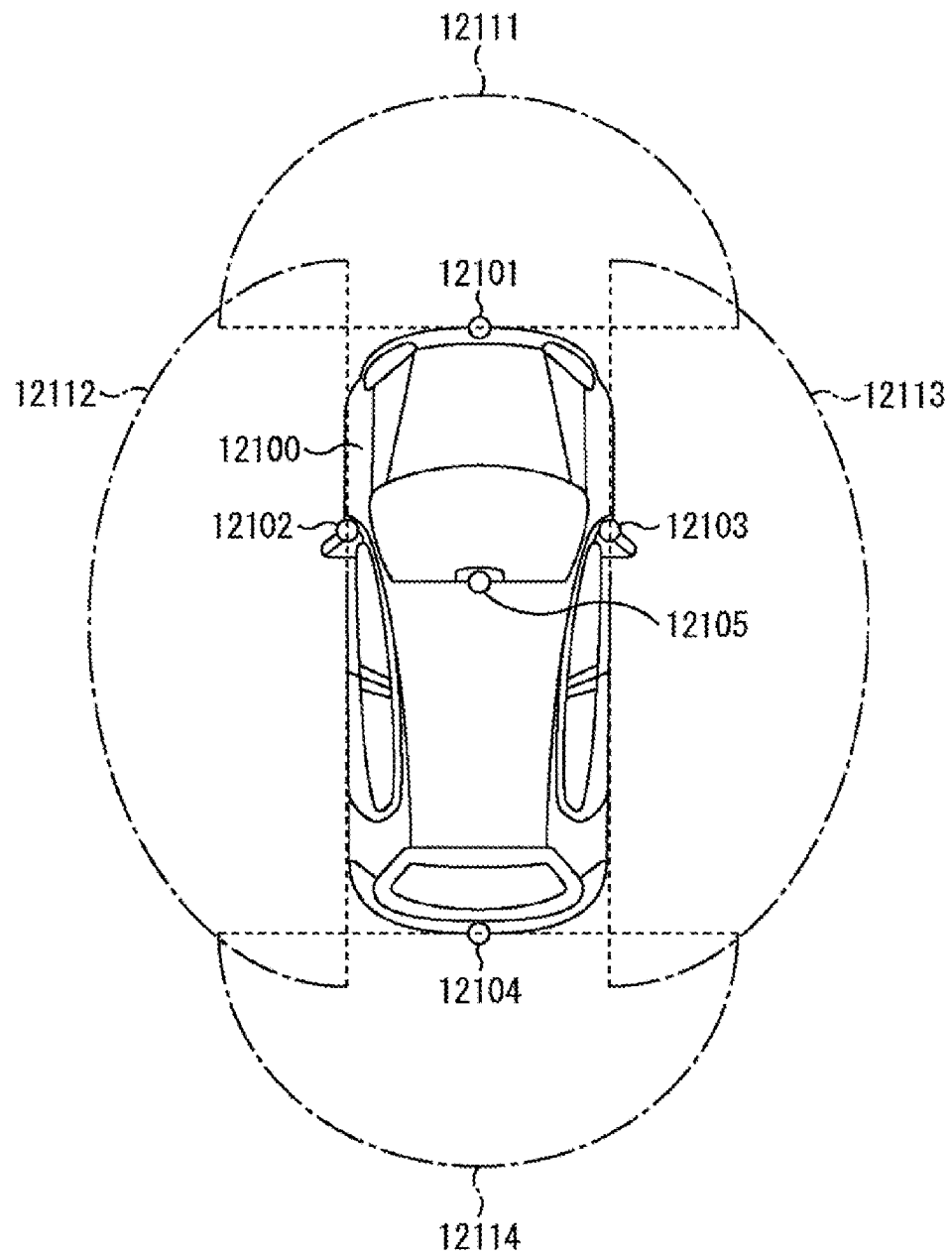


Fig. 28

Fig. 29



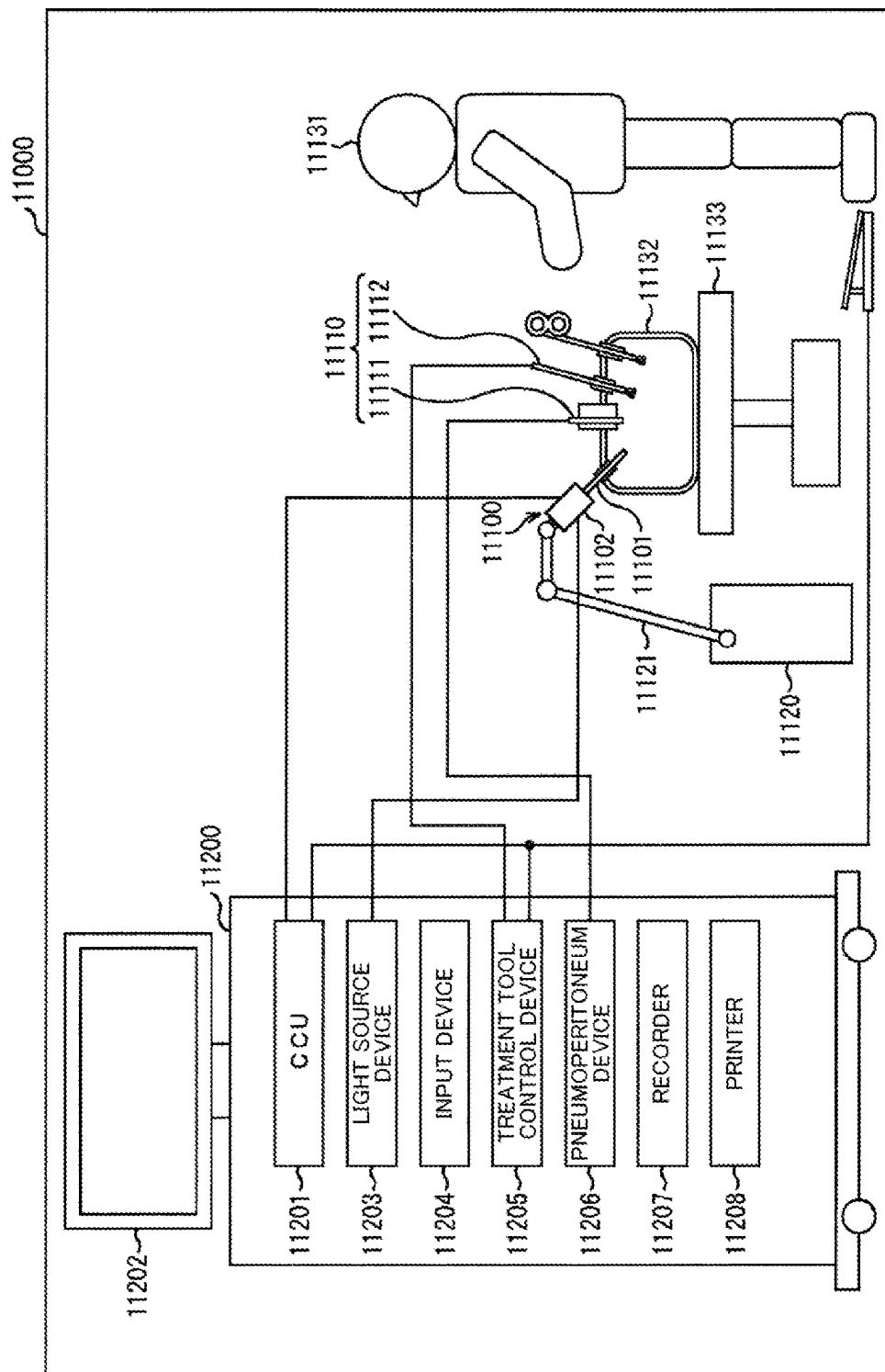
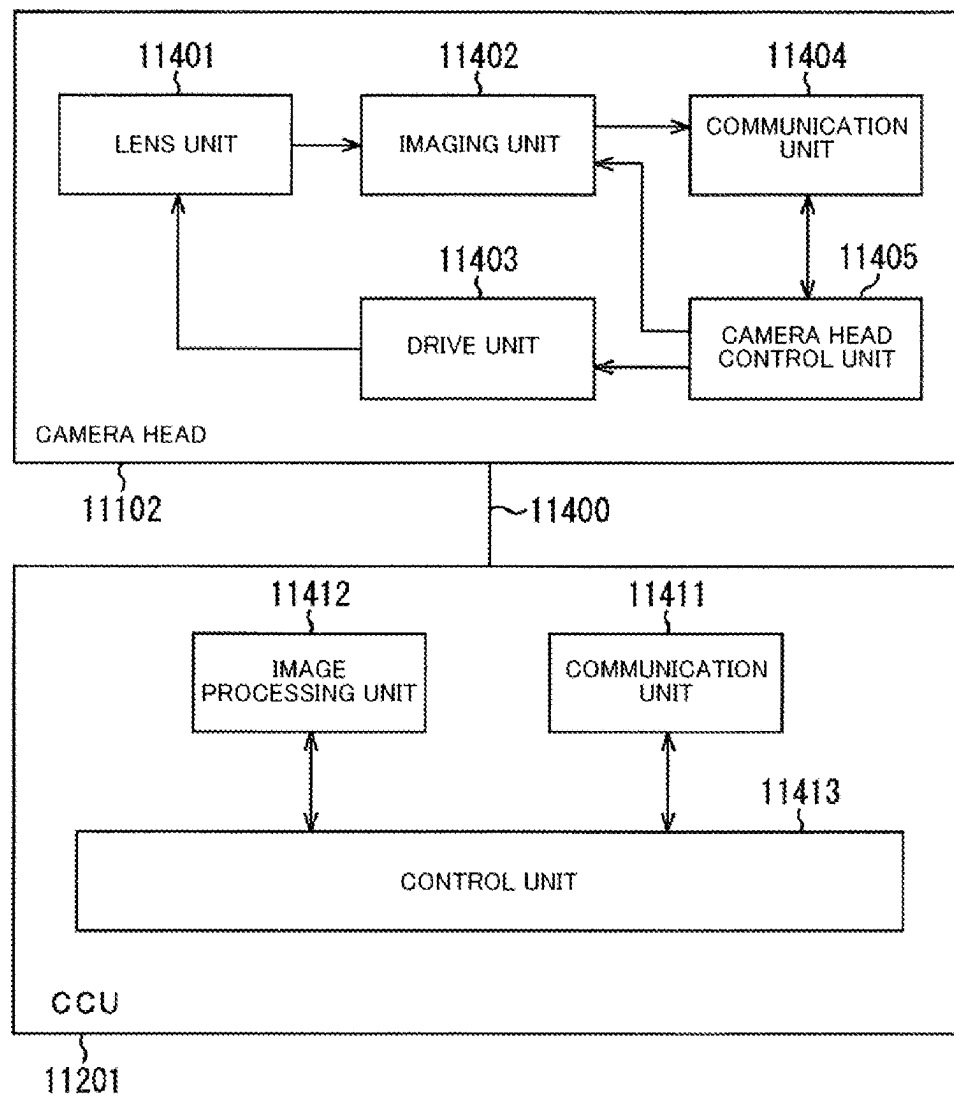


Fig. 30

Fig. 31



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SEMICONDUCTOR DEVICE, SOLID-STATE IMAGING DEVICE AND ELECTRONIC DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Phase of International Patent Application No. PCT/JP2020/037338 filed on Sep. 30, 2020, which claims priority benefit of Japanese Patent Application No. JP 2019-208612 filed in the Japan Patent Office on Nov. 19, 2019. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present technology relates to a semiconductor device, a solid-state imaging device and an electronic device.

BACKGROUND ART

In the related art, a semiconductor device including a first substrate and a second substrate bonded to the first substrate and in which, on a bonding surface between the first substrate and the second substrate, a first connection electrode formed on the first substrate and a second connection electrode formed on the second substrate are electrically connected has been proposed (for example, refer to PTL 1). In the semiconductor device described in PTL 1, an insulating film is arranged between the second connection electrode and a metal wiring provided in the second substrate to form a capacitor.

CITATION LIST

Patent Literature

[PTL 1]
JP 2011-166171 A

SUMMARY

Technical Problem

Incidentally, generally, when the first substrate and the second substrate are bonded, warp correction is performed on the first substrate and the second substrate, but stress is applied to the capacitor by the warp correction. In addition, stress is also applied to the capacitor due to the difference in the coefficient of thermal expansion between the second connection electrode and the insulating film. Here, in the semiconductor device described in PTL 1, since the position of the capacitor is close to the bonding surface and a contact area between the insulating film constituting the capacitor and the metal wiring is large, stress due to the bonding process is directly applied to the capacitor. Therefore, for example, insulating film cracks, variations in the polarization state of the insulating film, and the like may occur, and the performance of the capacitor may deteriorate.

An object of the present disclosure is to provide a semiconductor device, a solid-state imaging device and an electronic device through which it is possible to minimize deterioration of performance of a capacitor due to a bonding process.

Solution to Problem

A semiconductor device of the present disclosure includes (a) a first substrate, (b) a second substrate bonded to the first

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substrate, (c) a first electrode which is provided in the first substrate and of which one surface is positioned on the same surface as a bonding surface between the first substrate and the second substrate, (d) a second electrode which is provided in the second substrate and of which one surface is positioned on the same surface as the bonding surface and bonded to one surface of the first electrode, and (e) at least one of a first capacitor which is provided in the first substrate and of which one electrode is electrically connected to the other surface of the first electrode and a second capacitor which is provided in the second substrate and of which one electrode is electrically connected to the other surface of the second electrode.

A solid-state imaging device of the present disclosure includes a sensor substrate in which a plurality of photoelectric conversion units are arranged, a logic substrate which is bonded to the sensor substrate and in which a circuit that processes an electrical signal from the photoelectric conversion unit is integrated, a first electrode which is provided in the sensor substrate and of which one surface is positioned on the same surface as a bonding surface between the sensor substrate and the logic substrate, a second electrode which is provided in the logic substrate and of which one surface is positioned on the same surface as the bonding surface and bonded to one surface of the first electrode, and at least one of a first capacitor which is provided in the sensor substrate and of which one electrode is electrically connected to the other surface of the first electrode and a second capacitor which is provided in the logic substrate and of which one electrode is electrically connected to the other surface of the second electrode.

An electronic device of the present disclosure includes a solid-state imaging device including a sensor substrate in which a plurality of photoelectric conversion units are arranged, a logic substrate which is bonded to the sensor substrate and in which a circuit that processes an electrical signal from the photoelectric conversion unit is integrated, a first electrode which is provided in the sensor substrate and of which one surface is positioned on the same surface as a bonding surface between the sensor substrate and the logic substrate, a second electrode which is provided in the logic substrate and of which one surface is positioned on the same surface as the bonding surface and bonded to one surface of the first electrode, and at least one of a first capacitor which is provided in the sensor substrate and of which one electrode is electrically connected to the other surface of the first electrode and a second capacitor which is provided in the logic substrate and of which one electrode is electrically connected to the other surface of the second electrode, an optical lens that forms image light from a subject into an image on an imaging surface of the solid-state imaging device, and a signal processing circuit that performs signal processing on a signal output from the solid-state imaging device.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a cross-sectional view showing an overall structure of a semiconductor device according to a first embodiment.

FIG. 2 is a cross-sectional view showing an enlarged first electrode.

FIG. 3 is a cross-sectional view showing an enlarged first electrode.

FIG. 4 is a cross-sectional view showing an enlarged second electrode.

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FIG. 5 is a cross-sectional view showing an enlarged second electrode.

FIG. 6 is a cross-sectional view showing a first electrode and a second electrode bonded to each other.

FIG. 7 is a cross-sectional view showing a first electrode and a second electrode bonded to each other.

FIG. 8 is a cross-sectional view showing a first electrode and a second electrode bonded to each other.

FIG. 9 is a cross-sectional view showing a first electrode and a second electrode bonded to each other.

FIG. 10A is a cross-sectional view showing a first substrate before a region for substrate connection is formed.

FIG. 10B is a cross-sectional view showing a second substrate before a region for substrate connection is formed.

FIG. 11A is a cross-sectional view showing a first substrate after a region for substrate connection is formed.

FIG. 11B is a cross-sectional view showing a second substrate after a region for substrate connection is formed.

FIG. 12 is a cross-sectional view showing a first substrate and a second substrate after bonding.

FIG. 13 is a cross-sectional view showing a first substrate and a second substrate after thinning.

FIG. 14 is a plan view showing an overall structure of a sensor substrate.

FIG. 15 is a cross-sectional view showing a second electrode according to a modification example.

FIG. 16 is a cross-sectional view showing a second capacitor according to a modification example.

FIG. 17 is a cross-sectional view showing a region for substrate connection according to a modification example.

FIG. 18A is a cross-sectional view showing a second capacitor according to a modification example.

FIG. 18B is a plan view showing the second substrate in FIG. 18A when viewed from the bonding surface side.

FIG. 19 is a cross-sectional view showing a second capacitor according to a modification example.

FIG. 20 is a block diagram showing an example of a schematic configuration of a solid-state imaging device.

FIG. 21 is a cross-sectional view showing a cross section of a solid-state imaging device for one pixel.

FIG. 22 is a block diagram showing an example of a schematic configuration of a solid-state imaging device.

FIG. 23A is a cross-sectional view showing an enlarged first substrate.

FIG. 23B is a cross-sectional view showing an enlarged second substrate.

FIG. 24 is a circuit diagram showing a circuit configuration of a solid-state imaging device for one pixel.

FIG. 25 is a circuit diagram showing a circuit configuration of a solid-state imaging device for 4 pixels in two rows and two columns.

FIG. 26 is a timing chart illustrating a method of driving a solid-state imaging device.

FIG. 27 is a diagram showing an example of a schematic configuration of an electronic device.

FIG. 28 is a block diagram showing an example of a schematic configuration of a vehicle control system.

FIG. 29 is an explanatory diagram showing an example of installation positions of an outside-vehicle information detection unit and imaging units.

FIG. 30 is a diagram showing an example of a schematic configuration of an endoscopic surgery system.

FIG. 31 is a block diagram showing an example of a functional configuration of a camera head and a CCU.

DESCRIPTION OF EMBODIMENTS

Hereinafter, an example of a semiconductor device according to an embodiment of the present disclosure will be

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described with reference to FIG. 1 to FIG. 31. The embodiment of the present disclosure will be described in the following order. Here, the present disclosure is not limited to the following examples. In addition, the effects described in this specification are exemplary and not limiting, and other effects may be provided.

1. First embodiment: semiconductor device
 - 1-1 Configuration of semiconductor device
 - 1-2 Method of producing semiconductor device
 - 1-3 Modification example
2. Example of application to solid-state imaging device
 - 2-1 Configuration of solid-state imaging device
 - 2-2 Configuration of main part
 - 2-3 Modification example
3. Example of application to electronic device
4. Example of application to moving object
5. Example of application to endoscopic surgery system

1. First Embodiment

1-1 Configuration of Semiconductor Device

As shown in FIG. 1, a semiconductor device 1 according to a first embodiment includes a first substrate 2 and a second substrate 3 bonded to the first substrate 2. As the first substrate 2, for example, a sensor substrate in which a plurality of photoelectric conversion units are arranged can be used. In addition, as the second substrate 3, for example, a logic substrate in which circuits for processing electrical signals from photoelectric conversion units are integrated can be used. In addition, as a method of bonding the first substrate 2 to the second substrate 3, for example, plasma bonding can be used.

An interlayer insulating film 4 is provided on the first substrate 2 on the side of the second substrate 3 (in FIG. 1, the lower side). The interlayer insulating film 4 has a plurality of wiring layers therein. FIG. 1 illustrates an example in which four layers: a first wiring layer 5a, a second wiring layer 5b, a third wiring layer 5c and a fourth wiring layer 5d, are included as a plurality of wiring layers. That is, in FIG. 1, the interlayer insulating film 4, the first wiring layer 5a, the second wiring layer 5b, the third wiring layer 5c and the fourth wiring layer 5d form a multilayer wiring layer. Here, the number of wiring layers shown in FIG. 1 is an example, and the number of wiring layers included in the interlayer insulating film 4 may be a number other than 4.

Although details are omitted, each of the first wiring layer 5a, the second wiring layer 5b, the third wiring layer 5c and the fourth wiring layer 5d is electrically connected to an element or the like provided in the first substrate 2.

In addition, an interlayer insulating film 6 is provided on the front surface of the interlayer insulating film 4 on the side of the second substrate 3 (in FIG. 1, the lower side). As shown in FIG. 2 and FIG. 3, in the interlayer insulating film 6, a plurality of first electrodes 7 which are provided in the interlayer insulating film 6 and of which one surface (hereinafter referred to as an "exposed surface S1") is positioned on the same surface as a bonding surface S2 between the first substrate 2 and the second substrate 3 are arranged. That is, the exposed surface S1 of the first electrode 7 is exposed from the front surface (the bonding surface S2) of the interlayer insulating film 6 on the side of the second substrate 3. As the first electrode 7, for example, a copper electrode made of pure copper or copper alloy can be used.

As shown in FIG. 1, FIG. 2 and FIG. 3, the first electrode 7 includes a flat first electrode pad 8 of which one surface (in

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FIG. 1 to FIG. 3, the front surface on the lower side) is positioned on the same surface as the bonding surface S2 and a columnar first via 9 extending from the other surface (in FIG. 1 to FIG. 3, the front surface on the upper side) of the first electrode pad 8 in a thickness direction (in FIG. 1 and FIG. 2, the upward direction) of the first substrate 2.

In addition, as shown in FIG. 2, the wiring of the fourth wiring layer 5d, that is, the wiring of the wiring layer closest to the side of the second substrate 3 (in FIG. 2, the lower side) is electrically connected to the other surface (hereinafter referred to as a “non-exposed surface S3”) of a part of the first electrode 7 (hereinafter referred to as a “first electrode 7a”) among the plurality of first electrodes 7. In addition, as shown in FIG. 3, among the plurality of first electrodes 7, the wiring of the fourth wiring layer 5d is electrically connected to the other surface (the non-exposed surface S3) of a part of the first electrodes 7 (a part or all of the first electrodes 7 other than the first electrode 7a; hereinafter referred to as a “first electrode 7b”) with a first capacitor 10 provided in the interlayer insulating film 6 therebetween.

The first capacitor 10 includes two flat electrodes 11 and 12 that face each other and the insulating film 13 arranged between the two electrodes 11 and 12. The electrode 11 is arranged on the side of the second substrate 3 (in FIG. 3, the lower side), and the first electrode 7b is electrically connected thereto. In addition, the electrode 12 is arranged on the side opposite to the second substrate 3 side (in FIG. 3, the upper side), and the wiring of the fourth wiring layer 5d is electrically connected thereto. In addition, as the material of the electrodes 11 and 12, a conductor into which copper does not easily diffuse and which does not easily diffuse to the insulating film 13 is used. For example, high-melting-point metals such as tantalum (Ta), tantalum nitride (Ta_N), titanium (Ti), titanium nitride (Ti_N), tungsten nitride (W_N), zirconium nitride (Zr_N), and cobalt (Co), and high-melting-point metal nitrides may be exemplified. In addition, as the insulating film 13, for example, a single-layer film made of any of high-dielectric-constant insulating materials such as tantalum oxide (Ta₂O₅), hafnium oxide (HfO₂), aluminum oxide (Al₂O₃), silicon nitride (Si₃N₄) and zirconium oxide (ZrO₂), or a multilayer film formed by combining these is used.

Referring back to FIG. 1, an interlayer insulating film 14 is provided on the second substrate 3 on the side of the first substrate 2 (in FIG. 1, the upper side). The interlayer insulating film 14 has a plurality of wiring layers therein. FIG. 1 illustrates an example in which a first wiring layer 15a, a second wiring layer 15b, a third wiring layer 15c, a fourth wiring layer 15d, a fifth wiring layer 15e and a sixth wiring layer 15f are included as a plurality of wiring layers. That is, in FIG. 1, the interlayer insulating film 14 and the first wiring layer 15a to the sixth wiring layer 15f form a multilayer wiring layer. Here, the number of wiring layers shown in FIG. 1 is an example, and the number of wiring layers included in the interlayer insulating film 14 may be a number other than 6. Although details are omitted, each of the first wiring layer 15a to the sixth wiring layer 15f is electrically connected to an element or the like provided in the second substrate 3.

In addition, an interlayer insulating film 16 is provided on the front surface of the interlayer insulating film 14 on the side of the first substrate 2 (in FIG. 1, the upper side). As shown in FIG. 4 and FIG. 5, in the interlayer insulating film 16, a plurality of second electrodes 17 which are provided in the interlayer insulating film 16 and of which one surface (hereinafter referred to as an “exposed surface S4”) is

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positioned on the same surface as the bonding surface S2 are arranged. That is, the exposed surface S4 of the second electrodes 17 is exposed from the front surface (the bonding surface S2) of the interlayer insulating film 16 on the side of the first substrate 2. As the second electrode 17, for example, a copper electrode made of pure copper or copper alloy can be used.

As shown in FIG. 1, FIG. 4 and FIG. 5, the second electrode 17 includes a flat second electrode pad 18 of which one surface (in FIG. 1, FIG. 4 and FIG. 5, the front surface on the upper side) is positioned on the same surface as the bonding surface S2 and a columnar second via 19 extending from the other surface (in FIG. 1, FIG. 4 and FIG. 5, the front surface on the lower side) of the second electrode pad 18 in a thickness direction (in FIG. 1, FIG. 4 and FIG. 5, the downward direction) of the second substrate 3. The second electrode pad 18 is arranged on a surface object with the first electrode pad 8 and the bonding surface S2 interposed therebetween. That is, on the bonding surface S2, the first electrode pad 8 and the second electrode pad 18 are bonded to each other, and the interlayer insulating film 6 and the interlayer insulating film 16 are bonded to each other.

In addition, as shown in FIG. 4, the wiring of the sixth wiring layer 15f, that is, the wiring of the wiring layer closest to the side of the first substrate 2 (in FIG. 4, the upper side) is electrically connected to the other surface (hereinafter referred to as a “non-exposed surface S5”) of a part of the second electrode 17 (hereinafter referred to as a “second electrode 17a”) among the plurality of second electrodes 17. The second electrode 17a is connected to the first electrode 7a connected to the wiring of the fourth wiring layer 5d or the first electrode 7b connected to the wiring of the fourth wiring layer 5d with the first capacitor 10 therebetween. As shown in FIG. 6, at a connection part between the second electrode 17a and the first electrode 7a, the wiring of the fourth wiring layer 5d of the first substrate 2 and the wiring of the sixth wiring layer 15f of the second substrate 3 are electrically connected with the first electrode 7a and the second electrode 17a therebetween. Therefore, it is possible to transmit a DC signal or supply DC power between the wiring of the fourth wiring layer 5d of the first substrate 2 and the wiring of the sixth wiring layer 15f of the second substrate 3.

On the other hand, as shown in FIG. 7, at a connection part between the second electrode 17a and the first electrode 7b, the wiring of the fourth wiring layer 5d of the first substrate 2 and the wiring of the sixth wiring layer 15f of the second substrate 3 are electrically connected with the first capacitor 10 therebetween in addition to the first electrode 7b and the second electrode 17a. Therefore, the charge of these wirings can be accumulated in the first capacitor 10. When the configuration of the connection part shown in FIG. 7 is realized, for example, either the wiring of the fourth wiring layer 5d or the wiring of the sixth wiring layer 15f electrically connected to the first capacitor 10 may be used as a ground wiring.

In addition, as shown in FIG. 5, the wiring of the sixth wiring layer 15f is electrically connected to the other surface (the non-exposed surface S5) of a part of the second electrode 17 (a part or all of the second electrode 17 other than the second electrode 17a; hereinafter referred to as a “second electrode 17b”) among the plurality of second electrodes 17 with a second capacitor 20 provided in the interlayer insulating film 16 therebetween. Like the second electrode 17a, the second electrode 17b is connected to the first electrode 7a connected to the wiring of the fourth wiring layer 5d or the first electrode 7b connected to the wiring of

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the fourth wiring layer **5d** with the first capacitor **10** therebetween. As shown in FIG. **8**, at a connection part between the second electrode **17b** and the first electrode **7a**, the wiring of the fourth wiring layer **5d** of the first substrate **2** and the wiring of the sixth wiring layer **15f** of the second substrate **3** are electrically connected with the first electrode **7a**, the second electrode **17b** and the second capacitor **20** therebetween. Therefore, the charge of these wirings can be accumulated in the second capacitor **20**.

On the other hand, as shown in FIG. **9**, at a connection part between the second electrode **17b** and the first electrode **7b**, the wiring of the fourth wiring layer **5d** of the first substrate **2** and the wiring of the sixth wiring layer **15f** of the second substrate **3** are electrically connected with the first capacitor **10** therebetween in addition to the first electrode **7b**, the second electrode **17b** and the second capacitor **20**. Therefore, the charge of these wirings can be accumulated in the first capacitor **10** and the second capacitor **20**. Here, when the configuration of the connection part shown in FIG. **8** and FIG. **9** is realized, for example, either the wiring of the fourth wiring layer **5d** or the wiring of the sixth wiring layer **15f** electrically connected to the first capacitor **10** or the second capacitor **20** may be used as a ground wiring.

The second capacitor **20** includes two flat electrodes **21** and **22** that face each other and an insulating film **23** arranged between the two electrodes **21** and **22**. The electrode **21** is arranged on the side of the first substrate **2** (in FIG. **5**, the upper side), and the second electrode **17b** is electrically connected thereto. In addition, the electrode **22** is arranged on the side opposite to the first substrate **2** side (in FIG. **5**, the lower side), and the wiring of the sixth wiring layer **15f** is electrically connected thereto. In addition, as the material of the electrodes **21** and **22** and the insulating film **23**, for example, the same materials as those of the electrodes **11** and **12** and the insulating film **13** of the first substrate **2** can be used.

Here, the region between the fourth wiring layer **5d** of the first substrate **2** and the sixth wiring layer **15f** of the second substrate **3** is defined as a region for substrate connection **24**. That is, the region for substrate connection **24** is a region in which the interlayer insulating film **6** of the first substrate **2** and the interlayer insulating film **16** of the second substrate **3** are provided. Each of the first capacitor **10** and the second capacitor **20** is arranged on the outmost layer of the region for substrate connection **24**.

With such a configuration, as shown in FIG. **6**, in the region for substrate connection **24**, the wiring of the fourth wiring layer **5d** and the wiring of the sixth wiring layer **15f** are electrically connected with the first electrode **7a** and the second electrode **17a** therebetween, and thus a signal is transmitted and power is supplied between the first substrate **2** and the second substrate **3**. In addition, as shown in FIG. **7**, FIG. **8** and FIG. **9**, when the wiring of the fourth wiring layer **5d** and the wiring of the sixth wiring layer **15f** are electrically connected with the first capacitor **10** and the second capacitor **20** therebetween in addition to the first electrode **7b** and the second electrode **17a**, a capacitor function of accumulating the charge of the wiring of the fourth wiring layer **5d** and the charge of the wiring of the sixth wiring layer **15f** is realized.

1-2 Method of Producing Semiconductor Device

Next, a method of producing the semiconductor device **1** of the first embodiment will be described. FIGS. **10A**, **10B**,

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11A, **11B**, **12**, and **13** show diagrams illustrating a process of producing the semiconductor device **1** of the first embodiment.

First, as shown in FIGS. **10A** and **10B**, the first substrate **2** (for example, a sensor substrate) is prepared before the region for substrate connection **24**, a color filter **36**, and a micro lens **37** are formed and the second substrate **3** (for example, a logic substrate) is prepared before the region for substrate connection **24** is formed.

Subsequently, as shown in FIG. **11A**, the first capacitor **10**, the interlayer insulating film **6**, the first via **9** and the first electrode pad **8** are formed on the prepared first substrate **2** in that order. Thereby, the region for substrate connection **24** is formed on the first substrate **2**. Similarly, as shown in FIG. **11B**, the second capacitor **20**, the interlayer insulating film **16**, the second via **19** and the second electrode pad **18** are formed on the prepared second substrate **3** in that order. Thereby, the region for substrate connection **24** is formed on the second substrate **3**.

Subsequently, as shown in FIG. **12**, the first substrate **2** and the second substrate **3** are bonded so that the region for substrate connection **24** of the first substrate **2** and the region for substrate connection **24** of the second substrate **3** overlap each other. As a method of bonding the first substrate **2** to the second substrate **3**, for example, plasma bonding can be used.

Subsequently, as shown in FIG. **13**, the first substrate **2** is thinned.

Subsequently, as shown in FIG. **1**, the color filter **36**, the micro lens **37** and the like are formed on the light incident side of the first substrate **2**. Thereby, the semiconductor device **1** of the first embodiment is formed.

As described above, as shown in FIG. **1** to FIG. **5**, the semiconductor device **1** of the first embodiment includes at least one of the first capacitor **10** which is provided in the first substrate **2** and of which one electrode **11** is electrically connected to the non-exposed surface **S3** of the first electrode **7(7b)** and the second capacitor **20** which is provided in the second substrate **3** and of which one electrode **21** is electrically connected to the non-exposed surface **S5** of the second electrode **17(17b)**. Therefore, for example, compared with the method in which the first electrode **7(7b)** is used as the electrode **11** of the first capacitor **10**, and the second electrode **17(17b)** is used as the electrode **21** of the second capacitor **20**, since the positions of the first capacitor **10** and the second capacitor **20** are far from the bonding surface **S2**, the stress due to the bonding process is not directly applied to the first capacitor **10** and the second capacitor **20**. Therefore, it is possible to provide the semiconductor device **1** that can minimize deterioration of performance of the first capacitor **10** and the second capacitor **20** (capacitor) due to the bonding process.

In addition, since the first electrode **7(7b)** and the second electrode **17(17b)** do not come into contact with the insulating film **13** of the first capacitor **10** and the insulating film **23** of the second capacitor **20**, for example, even if the first electrode **7(7b)** and the second electrode **17(17b)** are a copper electrode made of pure copper or a copper alloy, it is possible to prevent copper contained in the first electrode **7(7b)** and the second electrode **17(17b)** from diffusing into the insulating films **13** and **23**. In addition, for example, compared with the method in which the first electrode **7** and the second electrode **17** are used as the electrodes **11** and **21** of the first capacitor **10** and the second capacitor **20**, it is possible to increase the degree of freedom for the material of the electrodes **11** and **21** and improve characteristics of the first capacitor **10** and the second capacitor **20**.

In addition, for example, unlike the method in which the first capacitor 10 and the second capacitor 20 are provided on the fourth wiring layer 5d of the first substrate 2 and the sixth wiring layer 15f of the second substrate 3, it is possible to prevent the first capacitor 10 from using the region in the fourth wiring layer 5d and the second capacitor 20 from using the region in the sixth wiring layer 15f, it is possible to minimize an increase in the size of the fourth wiring layer 5d and the sixth wiring layer 15f, and it is possible to minimize an increase in the size of the semiconductor device 1. In addition, it is possible to prevent the wiring density of the fourth wiring layer 5d and the sixth wiring layer 15f from increasing and improve the yield.

That is, in the semiconductor device 1 of the first embodiment, the fact that the region for substrate connection 24 between the first substrate 2 and the second substrate 3 is not a region in which the metal densities are closest as in the first wiring layer 5a to the fourth wiring layer 5d of the first substrate 2 and the first wiring layer 15a to the sixth wiring layer 15f of the second substrate 3 is focused on. According to the semiconductor device 1 of the first embodiment, when the first capacitor 10 and the second capacitor 20 are provided in the region for substrate connection 24, the region for substrate connection 24 can be effectively utilized.

In addition, in the semiconductor device 1 of the first embodiment, when the first capacitor 10 and the second capacitor 20 are used, it is possible to realize a function of cutting off noise generated in a power supply or an analog circuit and a storage function for storing charge temporarily. For example, when noise is contained in the supply of power, since the semiconductor device 1 operates with the supply of power containing noise, the semiconductor device 1 may not operate appropriately. On the other hand, when a noise cut-off function is realized, since the power from which noise has been removed is supplied to the semiconductor device 1, the semiconductor device 1 can be operated appropriately. In particular, when capacitors (the first capacitor 10 and the second capacitor 20) are mounted on the first substrate 2 and the second substrate 3, the noise removal performance can be doubled or better.

Here, a case in which a sensor substrate is used as the first substrate 2, a logic substrate is used as the second substrate 3, and a solid-state imaging device is configured as the semiconductor device 1 is conceivable. In such a solid-state imaging device (the semiconductor device 1), as shown in FIG. 14, a pixel region 25 in which a photoelectric conversion unit and the like are provided is arranged in the center of the chip of the sensor substrate (the first substrate 2), an I/O unit 26 in which an I/O pad and the like are provided is arranged on the outer peripheral part of the chip, and a driver 27, which is a transmission path for photoelectrically converted signals, is arranged between the pixel region 25 and the I/O unit 26. The driver 27 is formed by arranging the first electrode 7a (refer to FIG. 2) bonded to the second electrode 17a (refer to FIG. 4) in an array form. Therefore, the solid-state imaging device (the semiconductor device 1) transmits the signal photoelectrically converted in the pixel region 25 of the image sensor (the first substrate 2) to the logic substrate (the second substrate 3) through the driver 27, performs calculation in the logic substrate (the second substrate 3) based on the transmitted signal, and outputs the calculation result to an external display or the like through the I/O unit 26.

In the solid-state imaging device (the semiconductor device 1) shown in FIG. 14, generally, in a region other than the region occupied by the driver 27, that is, most of the

region of the sensor substrate (the first substrate 2), a dummy electrode for robustly bonding the first substrate 2 and the second substrate 3 is laid out. The dummy electrode is an electrode which includes an electrode pad, a via and the like as in the first electrode 7 and the second electrode 17, but is not electrically connected to either the first substrate 2 or the second substrate 3. Therefore, since the region in which the dummy electrode is laid out does not have a function for electrically connecting the first substrate 2 and the second substrate 3, it can be used for the layout of the first electrode 7a and the second electrode 17a of the first embodiment, that is, electrodes electrically connected to the first capacitor 10 and the second capacitor 20. Therefore, when the solid-state imaging device (the semiconductor device 1) shown in FIG. 14 is formed, a wide region can be prepared as a region in which the first capacitor 10 and the second capacitor 20 can be laid out, the degree of freedom for the layout of the first capacitor 10 and the second capacitor 20 can increase, and the difficulty in designing the solid-state imaging device (the semiconductor device 1) can be reduced.

In addition, in the solid-state imaging device (the semiconductor device 1) shown in FIG. 14, when noise is contained in the supply of power, since the solid-state imaging device (the semiconductor device 1) operates with the supply of power containing noise, the image produced by the solid-state imaging device (the semiconductor device 1) reacts sensitively to noise, and in a severe case, the image may be streaked. Here, the power supply is generated by an AC-DC conversion unit inside the system such as a smartphone, a camera, and a monitoring device in which the solid-state imaging device (the semiconductor device 1) is mounted, and supplied from a plurality of terminals provided in the I/O unit 26. On the other hand, when the solid-state imaging device (the semiconductor device 1) shown in FIG. 14 is formed, there is a wide region around the I/O unit 26 as a region in which the first capacitor 10 and the second capacitor 20 can be laid out. Therefore, a noise cut-off function can be realized by laying out the first capacitor 10 and the second capacitor 20 in the vicinity of the I/O unit 26, the power from which noise has been removed can be supplied to the solid-state imaging device (the semiconductor device 1), and a clearer image can be created by the solid-state imaging device (the semiconductor device 1).

In addition, when the first capacitor 10 and the second capacitor 20 are laid out in the vicinity of the I/O unit 26 and a noise cut-off function is realized, for example, noise can be removed from the signal output to the display via the I/O unit 26, an electrical load on the display can be reduced, and an image with little noise can be projected on the display.

Here, a case in which a function of cutting off noise at a macro level is realized has been described so far, but the same effect can be obtained for a case in which a function of cutting off noise at a micro level is realized. For example, in the solid-state imaging device (the semiconductor device 1) shown in FIG. 14, when the signal from the photoelectric conversion unit in an analog circuit such as a pixel transistor is superimposed with noise, even if it is digitized by an AD converter of pixel units or area units, it is difficult to prevent the signal quality from deteriorating. On the other hand, if a noise cut function can be realized in pixel units or area units using the first capacitor 10 and the second capacitor 20, it is possible to prevent the signal quality from deteriorating, and a clearer image can be created by the solid-state imaging device (the semiconductor device 1).

In addition, when a storage function is realized by laying out the first capacitor 10 and the second capacitor 20, it can

also be used for accumulating charges of the global shutter. By using it for charge accumulation, a fast-moving subject can be imaged with a distortion-free image.

In addition, in the semiconductor device 1 of the first embodiment, the first electrode 7 includes the first electrode pad 8 of which one surface is positioned on the same surface as the bonding surface S2 and the first via 9 extending from the other surface of the first electrode pad 8. In addition, the second electrode 17 includes the second electrode pad 18 of which one surface is positioned on the same surface as the bonding surface S2 and the second via 19 extending from the other surface of the second electrode pad 18. Therefore, the amount of copper used in the first electrode 7 and the second electrode 17 can be reduced, the production cost can be reduced, and the size of the bonding surface S2 between the first electrode 7 and the second electrode 17 can be increased.

In addition, in the semiconductor device 1 of the first embodiment, at least one of the first capacitor 10 and the second capacitor 20 includes two electrodes 11 and 12 that face each other and the insulating film 13 arranged between the two electrodes 11 and 12. Therefore, the first capacitor 10 and the second capacitor 20 can have a metal-insulator-metal (MIM) structure, and the first capacitor 10 and the second capacitor 20 can have a high capacitance density.

1-3 Modification Example

(1) Here, in the first embodiment, an example in which the second electrode 17 is composed of the second electrode pad 18 and the second via 19 is shown, but other configurations can be used. For example, as shown in FIG. 15, the second via 19 may be omitted, and the second electrode 17 may be composed of only the second electrode pad 18. Similarly, the first electrode 7 may be composed of only the first electrode pad 8.

(2) In addition, in the first embodiment, an example in which the electrode 22 of the second capacitor 20 is connected to the wiring of the sixth wiring layer 15f is shown, but other configurations can be used. For example, as shown in FIG. 16, electrical connection to a second electrode 17 different from the second electrode 17 connected to the electrode 21 may be formed. In FIG. 16, in order to prevent the electrode 22 and the wiring of the sixth wiring layer 15f from being electrically connected, an insulating film 28 is arranged therebetween. In addition, the periphery of the second via 19 of the second electrode 17 connected to the electrode 22 is covered with an insulating film 29 so that it is not electrically connected to the electrode 21. Similarly, the electrode 12 of the first capacitor 10 may be electrically connected to a first electrode 7 different from the first electrode 7 connected to the electrode 11.

(3) Here, generally, the capacitance C of the capacitor is defined by the following Formula (1) using a relative dielectric constant ϵ_r , a vacuum dielectric constant ϵ_0 , an electrode area S, and a distance d between electrodes.

$$C = \epsilon_r \cdot \epsilon_0 \cdot S / d \quad (1)$$

According to Formula (1), the capacitance C of the capacitor can be increased by any of reducing the distance d between electrodes, enlarging the electrode area S, and increasing the relative dielectric constant ϵ_r . However, in order to reduce the distance d between electrodes, the insulating film arranged between electrodes should be thinned, and there is a high possibility of short-circuiting between the electrodes. In particular, when capacitors are formed by stacking using a planar technology, in consider-

ation of flattening, a thin film of a material having a large relative dielectric constant ϵ_r is generally used as an insulating film so that a region in which the capacitors are formed is not higher than the other regions. Therefore, there is a higher possibility of short-circuiting between the electrodes. In addition, the probability of short-circuiting occurring between electrodes is largely influenced by defects of the insulating film caused by dust or the like, and is defined by the defect density of the insulating film. The defect density of the insulating film can be reduced to some extent by devising devices, processes and the like used for production of the semiconductor device 1, but cannot be reduced to zero.

On the other hand, a configuration in which at least one of the first capacitor 10 and the second capacitor 20 is divided and laid out using the high degree of freedom for the layout of the first capacitor 10 and the second capacitor 20 may be used. In this case, as shown in FIG. 17, a configuration in which two or more of the plurality of first capacitors 10 or two or more of the plurality of second capacitors 20 are connected to the same wiring is used. FIG. 17 illustrates a configuration in which two or more second capacitors 20 are connected to the same wiring (wiring of the fourth wiring layer 5d) in the first substrate 2. With the configuration shown in FIG. 17, two or more second capacitors 20 can be connected in parallel, and the same capacitance as a large capacitor having the same area as the total area of the second capacitors 20 connected in parallel can be realized. That is, since a large capacitance can be realized with a small capacitor (the second capacitor 20), it is possible to improve the yield of the semiconductor device 1. In addition, when the divided capacitor (the second capacitor 20) is redundantly laid out, it is possible to further improve the yield of the semiconductor device 1. Here, an example in which two or more of the plurality of second capacitors 20 are connected to the same wiring has been described, but the same effect can be obtained when two or more of the plurality of first capacitors 10 are connected to the same wiring.

(4) In addition, in the first embodiment, an example in which the first capacitor 10 has a configuration in which the insulating film 13 is arranged between the two electrodes 11 and 12 and the second capacitor 20 has a configuration in which the insulating film 23 is arranged between the two electrodes 21 and 22 is shown, but other configurations can be used. For example, as shown in FIGS. 18A and 18B, at least one of the first capacitor 10 and the second capacitor 20 may have a multilayer structure in which an electrode 30 and an insulating film 31 are alternately and repeatedly arranged.

FIGS. 18A and 18B illustrate a configuration in which the second capacitor 20 has a multilayer structure. In FIGS. 18A and 18B, the electrode 30 (hereinafter referred to as an "electrode 30a") closest to the side of the sixth wiring layer 15f among the electrodes 30 forming the multilayer structure is electrically connected to the sixth wiring layer 15f, and the other electrodes 30 (hereinafter referred to as an "electrode 30b," "electrode 30c," "electrode 30d," or "electrode 30e") electrically connected to the second electrodes 17 are different from each other. That is, each of the electrodes 30b to 30e corresponds to "one electrode 21" of the first embodiment. The periphery of the second via 19 of the second electrode 17 is covered with an insulating film 32 in order to prevent electrical connection to the electrode 30, which is not a connection target. In the second capacitor 20 shown in FIGS. 18A and 18B, a capacitor is formed by the electrodes 30a and 30b and the insulating film 31, a capacitor is formed by the electrodes 30b and 30c and the insulating film 31, a

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capacitor is formed by the electrodes **30c** and **30d** and the insulating film **31**, a capacitor is formed by the electrodes **30d** and **30e** and the insulating film **31**, and a total of four capacitors are formed.

With the configuration shown in FIGS. **18A** and **18B**, for example, the capacitor can be appropriately laid out even when there is a limitation on the region in which the capacitor can be laid out, for example, when a capacitor (the second capacitor **20**) is provided for each pixel. Here, an example in which the second capacitor **20** has a multilayer structure has been described, but the same effect can also be obtained when the first capacitor **10** has a multilayer structure.

(5) In addition, in the first embodiment, an example in which the first capacitor **10** and the second capacitor **20** have a flat shape is shown, but other configurations can be used. For example, as shown in FIG. **19**, the first capacitor **10** and the second capacitor **20** may have a 3D shape so that the areas of the electrodes **11** and **12**, **21**, **22** of the first capacitor **10** and the second capacitor **20** become large.

FIG. **19** illustrates a configuration in which the second capacitor **20** has a 3D shape. In FIG. **19**, the second capacitor **20** includes a bottom part **33** and side wall parts **34** and **35** extending from the edge part of the bottom part **33** in a direction intersecting the bottom part **33**. In addition, the second capacitor **20** is provided in the sixth wiring layer **15f**, and end surfaces of the side wall parts **34** and **35** on the side of the first substrate **2** are positioned on the same surface as the bonding surface **S2**. With the configuration shown in FIG. **19**, the area of the electrodes **21** and **22** of the second capacitor **20** can be increased, and the capacitance of the second capacitor **20** can be increased. In addition, when the second capacitors **20** having a large capacitance are connected in parallel, it is possible to realize the same capacitance as a capacitor having an area equal to the total area of the second capacitors **20** connected in parallel. In this case, when aluminum (Al) is used for the wiring of the sixth wiring layer **15f** in which the second capacitor **20** is provided, processing can be easily performed and the process cost can be reduced. Here, an example in which the second capacitor **20** is composed of the bottom part **33** and the side wall parts **34** and **35** has been described, but the same effect can also be obtained when the first capacitor **10** is composed of a bottom part and a side wall part.

2. Example of Application to Solid-State Imaging Device

2-1 Configuration of Solid-State Imaging Device

The technology according to the present disclosure (the present technology) can be applied to various products. For example, the technology according to the present disclosure may be applied to the solid-state imaging device.

FIG. **20** is a diagram showing an example of a schematic configuration of the solid-state imaging device to which the technology according to the present disclosure (the present technology) can be applied.

As shown in FIG. **20**, a solid-state imaging device **101** has a configuration including a pixel unit **103** composed of a plurality of pixels **102** arranged on a substrate **111** made of silicon, a vertical drive circuit **104**, a column signal processing circuit **105**, a horizontal drive circuit **106**, an output circuit **107**, a control circuit **108**, and the like.

A plurality of pixels **102** including a photoelectric conversion unit composed of a photodiode, a charge accumulated capacitance part, and a plurality of MOS transistors are

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regularly arranged on the substrate **111** in a 2D array form. The MOS transistors constituting the pixel **102** may be four MOS transistors composed of a transfer transistor, a reset transistor, a select transistor, and an amplifier transistor, or may be three MOS transistors excluding the select transistor.

The pixel unit **103** is composed of a plurality of pixels **102** that are regularly arranged in a 2D array form. The pixel unit **103** is composed of an effective pixel region in which light is actually received, and a signal charge generated by photoelectric conversion is amplified, and read in the column signal processing circuit **105**, and a black reference pixel region (not shown) for outputting optical black that serves as a reference for the black level. The black reference pixel region is generally formed on the outer peripheral part of the effective pixel region.

The control circuit **108** generates a clock signal, a control signal, or the like that serves as a reference for the operation of the vertical drive circuit **104**, the column signal processing circuit **105**, the horizontal drive circuit **106** and the like based on a vertical synchronization signal, a horizontal synchronization signal and a master clock. Therefore, the clock signal, the control signal and the like generated by the control circuit **108** are input to the vertical drive circuit **104**, the column signal processing circuit **105**, the horizontal drive circuit **106**, and the like.

The vertical drive circuit **104** is composed of, for example, shift registers, and sequentially selects and scans each pixel **102** of the pixel unit **103** in row units in the vertical direction. Therefore, the pixel signal based on the signal charge generated in the photodiode of each pixel **102** according to the intensity of the light received is supplied to the column signal processing circuit **105** through a vertical signal line **109**.

The column signal processing circuit **105** is arranged, for example, for each column of the pixel **102**, and the signal output from the pixel **102** for one row is subjected to signal processing such as noise removal and signal amplification using the signal from the black reference pixel region (not shown, but formed around the effective pixel region) for each pixel column. A horizontal selection switch (not shown) is provided between the output end of the column signal processing circuit **105** and a horizontal signal line **110**.

The horizontal drive circuit **106** is composed of, for example, shift registers, and sequentially outputs a horizontal scanning pulse and thus selects each of the column signal processing circuits **105** in order, and outputs a pixel signal from each of the column signal processing circuits **105** to the horizontal signal line **110**.

The output circuit **107** performs signal processing on the signal sequentially supplied from each of the column signal processing circuits **105** through the horizontal signal line **110** and outputs it.

2-2 Configuration of Sensor Substrate

Next, a schematic configuration of the sensor substrate constituting the solid-state imaging device **101** will be described. FIG. **21** is a cross-sectional view showing a cross section of a sensor substrate **20000** for one pixel.

As shown in FIG. **21**, in the sensor substrate **20000**, a photodiode (PD) **20019** receives incident light **20001** that enters from the back surface (the top surface in FIG. **21**) of a semiconductor substrate **20018**. A flattening film **20013**, a color filter (CF) **20012**, and a micro lens **20011** are provided above the PD **20019**, and the incident light **20001** incident

sequentially through respective units is received on a light receiving surface **20017** and subjected to photoelectric conversion.

For example, in the PD **20019**, an n-type semiconductor region **20020** is formed as a charge accumulated region in which charges (electrons) are accumulated. In the PD **20019**, the n-type semiconductor region **20020** is provided inside p-type semiconductor regions **20016** and **20041** of the semiconductor substrate **20018**. The p-type semiconductor region **20041** having a higher impurity concentration than the back surface (top surface) side is provided on the front surface (bottom surface) side of the semiconductor substrate **20018** of the n-type semiconductor region **20020**. That is, the PD **20019** has a hole-accumulation diode (HAD) structure, and the p-type semiconductor regions **20016** and **20041** are formed at each interface between the top surface side and the bottom surface side of the n-type semiconductor region **20020** so that generation of a dark current is restricted.

A pixel separation unit **20030** that electrically separates a plurality of pixels **20010** is provided inside the semiconductor substrate **20018**, and the PD **20019** is provided in a region partitioned by the pixel separation unit **20030**. In the drawing, when the solid-state imaging device is viewed from the top surface side, for example, the pixel separation unit **20030** is formed in a grid pattern between the plurality of pixels **20010**, and the PD **20019** is formed in the region partitioned by the pixel separation unit **20030**.

In each PD **20019**, the anode is grounded, and in the solid-state imaging device, the signal charge (for example, electron) accumulated in the PD **20019** is read via a transfer Tr (MOS FET) (not shown) or the like, and output as an electrical signal to a vertical signal line (VSL) (not shown).

A wiring layer **20050** is provided on the front surface (bottom surface) of the semiconductor substrate **20018** on the side opposite to the back surface (top surface) in which respective units such as a light-blocking film **20014**, the CF **20012**, and the micro lens **20011** are provided.

The wiring layer **20050** includes a wiring **20051** and an insulation layer **20052**, and is formed so that the wiring **20051** is electrically connected to each element in the insulation layer **20052**. The wiring layer **20050** is a layer of so-called multilayer wiring, and is formed by alternately laminating the interlayer insulating film constituting the insulation layer **20052** and the wiring **20051** a plurality of times. Here, as the wiring **20051**, the wiring to the Tr for reading the charge from the PD **20019** such as the transfer Tr and each wiring such as VSL are laminated with the insulation layer **20052** therebetween.

A region for substrate connection **20061** is provided on the surface of the wiring layer **20050** opposite to the side on which the PD **20019** is provided. The region for substrate connection **20061** includes a capacitor **20062**, an electrode **20063**, and an interlayer insulating film **20064**, and in the interlayer insulating film **20064**, one electrode of the capacitor **20062** is electrically connected to the wiring **20051** of the wiring layer **20050**. In addition, the other electrode of the capacitor **20062** is electrically connected to the electrode **20063**, and the electrode **20063** is exposed from the back surface (the bottom surface in the drawing) of the region for substrate connection **20061**. FIG. 21 illustrates the electrode **20063** including a flat electrode pad **20065** positioned on the same surface as the back surface of the region for substrate connection **20061** and a columnar via **20066** extending from the back surface (the top surface in the drawing) of the electrode pad **20065** in the thickness direction (the upward direction in FIG. 21) of the sensor substrate **20000**. The region for substrate connection **20061** is bonded to a region

for substrate connection (notshown) of a logic substrate so that the electrode (not shown) of the logic chip and the electrode **20063** overlap each other.

The light-blocking film **20014** is provided on the side of the back surface (the top surface in the drawing) of the semiconductor substrate **20018**.

The light-blocking film **20014** is configured to block some of the incident light **20001** that is directed toward the back surface of the semiconductor substrate **20018** from above the semiconductor substrate **20018**.

The light-blocking film **20014** is provided above the pixel separation unit **20030** provided inside the semiconductor substrate **20018**. Here, the light-blocking film **20014** is provided on the back surface (the top surface) of the semiconductor substrate **20018** so that it protrudes in a convex shape with an insulating film **20015** such as a silicon oxide film therebetween. On the other hand, above the PD **20019** provided inside the semiconductor substrate **20018**, the light-blocking film **20014** is not provided and is open so that the incident light **20001** is incident on the PD **20019**.

That is, in the drawing, when the solid-state imaging device is viewed from the top surface side, the planar shape of the light-blocking film **20014** is a grid pattern, and an opening through which the incident light **20001** passes to the light receiving surface **20017** is formed.

The light-blocking film **20014** is formed of a light-blocking material that blocks light. For example, the light-blocking film **20014** is formed by sequentially laminating a titanium (Ti) film and a tungsten (W) film. In addition to this, the light-blocking film **20014** can be formed by, for example, sequentially laminating a titanium nitride (TiN) film and a tungsten (W) film.

The light-blocking film **20014** is covered with the flattening film **20013**. The flattening film **20013** is formed using an insulating material through which light passes.

The pixel separation unit **20030** includes a groove part **20031**, a fixed charge film **20032**, and an insulating film **20033**.

The fixed charge film **20032** is formed on the side of the back surface (top surface) of the semiconductor substrate **20018** so that it covers the groove part **20031** that partitions between the plurality of pixels **20010**.

Specifically, the fixed charge film **20032** is provided so that it covers the inner surface of the groove part **20031** formed on the side of the back surface (top surface) in the semiconductor substrate **20018** with a certain thickness. Therefore, the insulating film **20033** is provided (filled) to embed the inside of the groove part **20031** covered with the fixed charge film **20032**.

Here, the fixed charge film **20032** is formed using a high dielectric material having a negative fixed charge so that a positive charge (hole) accumulated region is formed at the interface part with the semiconductor substrate **20018** and generation of a dark current is restricted. When the fixed charge film **20032** is formed so that it has a negative fixed charge, an electric field is applied to the interface with the semiconductor substrate **20018** according to the negative fixed charge, and a positive charge (hole) accumulated region is formed.

The fixed charge film **20032** can be formed as, for example, a hafnium oxide film (HfO₂ film). In addition, for example, the fixed charge film **20032** can be formed so that it contains at least one other oxide of hafnium, zirconium, aluminum, tantalum, titanium, magnesium, yttrium, and the lanthanoid elements.

An example of the solid-state imaging device to which the technology according to the present disclosure can be

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applied has been described above. The technology according to the present disclosure can be applied to the sensor substrate **20000** within the configuration described above. Specifically, the first capacitor **10** in FIG. **1** can be applied to the capacitor **20062** in FIG. **21**, and the first electrode **7** in FIG. **1** can be applied to the electrode **20063** in FIG. **21**. When the technology according to the present disclosure is applied to the sensor substrate **20000**, since deterioration of the performance of the capacitor due to the bonding process can be minimized, a better captured image can be obtained.

2-3 Modification Example

In addition, the present technology may be applied to a solid-state imaging device having a global shutter function. An overall structure of the solid-state imaging device according to this modification example is not shown because it is the same as in FIG. **20**. FIG. **22** is a cross-sectional view of the solid-state imaging device **101** according to this modification example for one pixel.

As shown in FIG. **22**, the solid-state imaging device **101** includes a first substrate **180** on which a photoelectric conversion unit PD is formed and a second substrate **181** on which a charge accumulated capacitance part **153** and a plurality of MOS transistors are formed. Therefore, the first substrate **180** and the second substrate **181** are laminated and bonded. In addition, the side of the first substrate **180** on which the photoelectric conversion unit PD is formed constitutes the light incident surface on which light **L** is incident, and a color filter **159** and an on-chip lens **160** are formed on the light incident surface of the first substrate **180**.

The configuration of the first substrate **180** and the second substrate **181** will be described in detail with reference to FIGS. **23A** and **23B**.

First, the first substrate **180** will be described.

As shown in FIG. **23A**, the first substrate **180** is composed of a photoelectric conversion unit PD, a semiconductor substrate **112** in which an impurity region **116**, which is a drain of a first transfer transistor Tr1, is formed, a multilayer wiring layer **117** formed on the semiconductor substrate **112**, and a region for substrate connection **190** formed on the multilayer wiring layer **117**.

The semiconductor substrate **112** is formed of an N-type silicon substrate, and a P-type well layer **113** is formed on the semiconductor substrate **112**. The P-type well layer **113** can be formed by ion implanting P-type impurities into the semiconductor substrate **112**.

The photoelectric conversion unit PD is composed of an N-type well layer **114** formed on the P-type well layer **113**, and a P+ type impurity region **115** which is a region in contact with the N-type well layer **114** and formed on the front surface side of the P-type well layer **113**. The N-type well layer **114** is formed by ion implanting N-type impurities into a desired region of the P-type well layer **113**. In addition, the P+ type impurity region **115** is formed by ion implanting P-type impurities at a high concentration in a desired region of the P-type well layer **113**.

In the photoelectric conversion unit PD, a hole accumulation diode (HAD: registered trademark) structure is formed according to the effects of the pn junction between the P+ type impurity region **115** and the N-type well layer **114**, and the pn junction between the N-type well layer **114** and the P-type well layer **113**.

In the photoelectric conversion unit PD having such a configuration, a signal charge corresponding to the light intensity of incident light **L** is generated, and the photoelectrically converted signal charge is accumulated in the deple-

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tion layer formed between the P+ type impurity region **115** and the N-type well layer **114**.

The impurity region **116** is formed on the front surface side of the P-type well layer **113** in a region separated from the photoelectric conversion unit PD by a predetermined distance, and is a region in which the signal charge transferred from the photoelectric conversion unit PD is temporarily accumulated. The impurity region **116** is formed by ion implanting N-type impurities at a high concentration in a desired region of the P-type well layer **113**.

In this modification example, the region between the photoelectric conversion unit PD and the impurity region **116** is defined as a channel unit of the first transfer transistor Tr1.

The multilayer wiring layer **117** is formed on the P-type well layer **113** in which the photoelectric conversion unit PD and the impurity region **116** are formed, on the semiconductor substrate **112**. In the multilayer wiring layer **117**, a gate electrode **119** constituting the first transfer transistor Tr1 and a first wiring layer M1 formed on the gate electrode **119** are laminated with an interlayer insulating film **118** therebetween.

The gate electrode **119** is formed on the channel unit between the photoelectric conversion unit PD and the impurity region **116** formed in the P-type well layer **113** with a gate insulating film (not shown) therebetween.

In the first wiring layer M1, a first connection wiring **123** and a second connection wiring **122** are formed. The first connection wiring **123** is connected to the impurity region **116** with a contact part **121** formed in the interlayer insulating film **118** therebetween. In addition, the second connection wiring **122** is connected to the gate electrode **119** with a contact part **120** therebetween formed in the interlayer insulating film **118**.

The region for substrate connection **190** is formed on the first wiring layer M1 in the multilayer wiring layer **117**. In the region for substrate connection **190**, a second wiring layer M2 is laminated on the first wiring layer M1 with an interlayer insulating film **191** therebetween.

In the second wiring layer M2, a first connection electrode **127** and a second connection electrode **126** are formed, and the first connection electrode **127** and the second connection electrode **126** are formed to be exposed at the front surface of the multilayer wiring layer **117**. The first connection electrode **127** is connected to the first connection wiring **123** formed of the first wiring layer M1 with a contact part **124** formed in the interlayer insulating film **191**. In addition, the second connection electrode **126** is connected to the second connection wiring **122** formed of the first wiring layer M1 with a contact part **125** formed in the interlayer insulating film **191** therebetween.

In the first substrate **180** having the above configuration, the side opposite to the side of the semiconductor substrate **112** on which the first connection electrode **127** and the second connection electrode **126** are formed is the light incident side.

Next, the second substrate **181** will be described.

As shown in FIG. **23B**, the second substrate **181** is composed of a semiconductor substrate **128** on which impurity regions **130**, **131**, **132**, **134**, and **135**, which are sources/drains of a plurality of MOS transistors, are formed, a multilayer wiring layer **136** formed on the semiconductor substrate **128**, and a region for substrate connection **195** formed on the multilayer wiring layer **136**. Therefore, the charge accumulated capacitance part **153** is formed in the region for substrate connection **195**. In this modification example, the plurality of MOS transistors formed in the

second substrate **181** are a second transfer transistor **Tr2**, a reset transistor **Tr3**, an amplifying transistor **Tr4**, and a select transistor **Tr5**.

The semiconductor substrate **128** is formed of an N-type silicon substrate, and a P-type well layer **129** is formed on the semiconductor substrate **128**. The P-type well layer **129** can be formed by ion implanting P-type impurities into the semiconductor substrate **128**.

The impurity regions **130**, **131**, **132**, **134**, and **135** constituting the second transfer transistor **Tr2**, the reset transistor **Tr3**, the amplifying transistor **Tr4**, and the select transistor **Tr5** are formed in desired regions on the front surface side of the P-type well layer **129**. These impurity regions **130**, **131**, **132**, **134**, and **135** are formed by ion implanting N-type impurities at a high concentration in a desired region of the P-type well layer **129**.

The impurity region **130** is used as the source of the second transfer transistor **Tr2**. In addition, the impurity region **131** is shared by the drain of the second transfer transistor **Tr2** and the source of the reset transistor **Tr3**, and is used as a floating diffusion region in which the signal charge is read. In addition, the impurity region **132** shared by the drain of the reset transistor **Tr3** and the source of the amplifying transistor **Tr4**. In addition, the impurity region **134** is shared by the drain of the amplifying transistor **Tr4** and the source of the select transistor **Tr5**. In addition, the impurity region **135** serves as the drain of the select transistor **Tr5**. Therefore, the region of the P-type well layer **129** between the impurity regions **130**, **131**, **132**, **134**, and **135** is used as a channel unit constituting each MOS transistor.

The multilayer wiring layer **136** is formed on the P-type well layer **129** in which the impurity regions **130**, **131**, **132**, **134**, and **135** are formed, on the semiconductor substrate **128**. In the multilayer wiring layer **136**, gate electrodes **138**, **139**, **140**, and **141** constituting MOS transistors, a first wiring layer **M1'** and a second wiring layer **M2'** are laminated with an interlayer insulating film **137** therebetween.

The gate electrodes **138**, **139**, **140**, and **141** are formed on the channel unit constituting each MOS transistor with a gate insulating film (not shown) therebetween. The gate electrode **138** formed on the P-type well layer **129** between the impurity region **130** and the impurity region **131** is the gate electrode **138** of the second transfer transistor **Tr2**. In addition, the gate electrode **139** formed on the P-type well layer **129** between the impurity region **131** and the impurity region **132** is used as the gate electrode of the reset transistor **Tr3**. In addition, the gate electrode **140** formed on the P-type well layer **129** between the impurity region **132** and the impurity region **134** is used as the gate electrode of the amplifying transistor **Tr4**. In addition, the gate electrode **141** formed on the P-type well layer **129** between the impurity region **134** and the impurity region **135** is used as the gate electrode of the select transistor **Tr5**.

The first wiring layer **M1'** is formed on the gate electrodes **138**, **139**, **140**, and **141** with the interlayer insulating film **137** therebetween, and in the first wiring layer **M1'**, a first connection wiring **150**, a second connection wiring **149**, a selection wiring **148**, and the vertical signal line **109** (refer to FIG. 20) are formed. The first connection wiring **150** is connected to the impurity region **130** as the source of the second transfer transistor **Tr2** with a contact part **142** formed in the interlayer insulating film **137** therebetween. The second connection wiring **149** is connected to the impurity region **131** and the gate electrode **140** of the amplifying transistor **Tr4** with contact parts **143** and **144** formed in the interlayer insulating film **137** therebetween. That is, the impurity region **131**, which is a floating diffusion region, and

the gate electrode **140** of the amplifying transistor **Tr4** are electrically connected by the second connection wiring **149**. In addition, the selection wiring **148** is connected to the gate electrode **141** of the select transistor **Tr5** with a contact part **145** formed in the interlayer insulating film **137** therebetween. Therefore, a selection pulse is supplied from the selection wiring **148** to the gate electrode **141** of the select transistor **Tr5**. In addition, the vertical signal line **109** (refer to FIG. 20) is connected to the impurity region **135**, which is the drain of the select transistor **Tr5**, with a contact part **146** formed in the interlayer insulating film **137**.

In the second wiring layer **M2'**, a third connection wiring **152** and a fourth connection wiring **151** are formed. The third connection wiring **152** is connected to the first connection wiring **150** with a contact part **147** formed in the interlayer insulating film **137**. In addition, the fourth connection wiring **151** is formed to extend to a predetermined region. In addition, a first transfer wiring formed in the multilayer wiring layer **136** of the second substrate **181** (not shown) is connected to the fourth connection wiring **151**, and a first transfer pulse is supplied from the first transfer wiring to the fourth connection wiring **151**.

The region for substrate connection **195** is formed on the third connection wiring **152** and the fourth connection wiring **151**, in the multilayer wiring layer **136**. In the region for substrate connection **195**, the charge accumulated capacitance part **153** is formed on the third connection wiring **152** and the fourth connection wiring **151** of the second wiring layer **M2'**, and a third wiring layer **M3'** is formed on the second wiring layer **M2'** with the charge accumulated capacitance part **153** therebetween. That is, the charge accumulated capacitance part **153** is interposed between the second wiring layer **M2'** and the third wiring layer **M3'**. As the charge accumulated capacitance part **153**, a capacitor having an MIM structure can be used.

In the third wiring layer **M3'**, a first connection electrode **156** and a second connection electrode **157** are formed, and the first connection electrode **156** and the second connection electrode **157** are formed to be exposed at the front surface of the region for substrate connection **195**. The first connection electrode **156** is connected to the third connection wiring **152** formed of the second wiring layer **M2'** with a contact part **155** formed in an interlayer insulating film **196** therebetween, and is formed so that it extends on the fourth connection wiring **151** formed of the second wiring layer **M2'** with the charge accumulated capacitance part **153** therebetween. In addition, the second connection electrode **157** is connected to the fourth connection wiring **151** formed of the second wiring layer **M2'** with a contact part **154** formed in the interlayer insulating film **196** therebetween.

Here, although not shown in FIG. 23B, a second transfer wiring for supplying a second transfer pulse is connected to the gate electrode **138** of the second transfer transistor **Tr2**. Similarly, a reset wiring for supplying a reset pulse is also connected to the gate electrode **139** of the reset transistor **Tr3**. Therefore, the second transfer wirings and the reset wiring are formed by a desired wiring layer formed in the multilayer wiring layer **136**.

Therefore, the solid-state imaging device **101** of this modification example is formed by laminating the first substrate **180** on the second substrate **181** so that the first connection electrodes **156** and **127**, and the second connection electrodes **157** and **126** of the first substrate **180** and the second substrate **181** are connected to each other. Therefore, when the first substrate **180** and the second substrate **181** are bonded, the impurity region **116** constituting the first transfer transistor **Tr1**, the charge accumulated capacitance part

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153, and the impurity region 130 constituting the second transfer transistor Tr2 are electrically connected. In addition, in the solid-state imaging device 101 of this modification example, when the first substrate 180 and the second substrate 181 are laminated and bonded, the photoelectric conversion unit PD and the charge accumulated capacitance part 153 are three-dimensionally laminated.

In addition, in the solid-state imaging device 101 of this modification example, the first connection electrode 156 serves as a light-blocking film, and the impurity region 130 as the source of the second transfer transistor Tr2 is shielded from light by the first connection electrode 156. Therefore, since the amount of light incident on the impurity region 130 is reduced, and generation of unnecessary signal charges is minimized, color mixing is reduced. Thus, in this case, all regions except the opening of the photoelectric conversion unit PD are preferably shielded from light.

Next, a method of driving the solid-state imaging device 101 of this modification example will be described with reference to FIG. 24. FIG. 24 is a circuit configuration of the solid-state imaging device 101 of this modification example for one pixel, and FIG. 25 is a circuit configuration for 4 pixels in two rows and two columns adjacent to each other.

The line a in FIG. 24 indicates an electrode connection surface between the first connection electrode 127 and the second connection electrode 126 formed in the first substrate 180, and the first connection electrode 156 and the second connection electrode 157 formed in the second substrate 181.

The side of the anode of the photodiode, which is a photoelectric conversion unit PD, is grounded, and the side of the cathode is connected to the source of the first transfer transistor Tr1. In addition, although not shown in FIG. 22, as shown in FIG. 24 and FIG. 25, in the first substrate 180, a reset transistor for a photoelectric conversion unit Tr6 is formed, and the drain of the reset transistor for a photoelectric conversion unit Tr6 is connected to the side of the cathode of the photoelectric conversion unit PD. A power supply voltage wiring 185 for applying a power supply voltage VDD is connected to the source of the reset transistor for a photoelectric conversion unit Tr6. In addition, a reset wiring 175 for supplying a reset pulse ϕ PDRST is connected to a gate electrode 162 of the reset transistor for a photoelectric conversion unit Tr6.

The drain of the first transfer transistor Tr1 is connected to the source of the second transfer transistor Tr2 with the first connection electrode 156 connected to one electrode of the charge accumulated capacitance part 153 therebetween. A first transfer wiring 184 for supplying a first transfer pulse ϕ TRG1 is connected to the gate electrode 119 of the first transfer transistor Tr1. In addition, the first transfer wiring 184 is connected to the fourth connection wiring 151 connected to the other electrode of the charge accumulated capacitance part 153.

The drain of the second transfer transistor Tr2 is connected to the source of the reset transistor Tr3, and also connected to the gate electrode 140 of the amplifying transistor Tr4. A second transfer wiring 163 for supplying a second transfer pulse ϕ TRG2 is connected to the gate electrode 138 of the second transfer transistor Tr2.

A power supply voltage wiring 188 for applying a power supply voltage VDD is connected to the drain of the reset transistor Tr3, and a reset wiring 164 for supplying a reset pulse ϕ RST is connected to the gate electrode 139 of the reset transistor Tr3.

The power supply voltage wiring 188 for applying a power supply voltage VDD is connected to the source of the

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amplifying transistor Tr4, and the drain of the amplifying transistor Tr4 is connected to the source of the select transistor Tr5.

The selection wiring 148 for supplying a selection pulse ϕ SEL is connected to the gate electrode 141 of the select transistor Tr5, and the drain of the select transistor Tr5 is connected to the vertical signal line 109 (refer to FIG. 20).

Therefore, as shown in FIG. 25, in the solid-state imaging device 101 in which the pixels 102 are arranged in a 2D matrix form, the second transfer wiring 163, the reset wiring 164, and the selection wiring 148 common to each row are connected to the gate electrodes 138, 139, and 141. Therefore, the second transfer pulse ϕ TRG2, the reset pulse ϕ RST, and the selection pulse ϕ SEL input to the gate electrodes 138, 139, and 141 are supplied from the vertical drive circuit 104. Although not shown, the reset pulse ϕ PDRST supplied to the gate electrode 162 of the reset transistor for a photoelectric conversion unit Tr6 and the first transfer pulse ϕ TRG1 supplied to the gate electrode 119 of the first transfer transistor Tr1 are also supplied by the vertical drive circuit 104.

In addition, the vertical signal line 109 (refer to FIG. 20) common to each column is connected to the drain of the select transistor Tr5.

The column signal processing circuit 105 provided for each column is connected to a stage after the vertical signal line 109. Therefore, a horizontal transistor Tr7 to which a horizontal selection pulse from the horizontal drive circuit 106 is input is connected to a stage after the column signal processing circuit 105.

Next, a method of driving the solid-state imaging device 101 having the above circuit configuration will be described using the timing chart shown in FIG. 26 and the circuit configuration of FIG. 25.

First, the reset pulse ϕ PDRST is set to high, and the reset transistors for a photoelectric conversion unit Tr6 of all pixels are turned on at the same time, and thus the potential of the photoelectric conversion unit PD of all pixels is reset to the same potential as the power supply voltage VDD. That is, according to this operation, unnecessary charges stored in the photoelectric conversion unit PD of all pixels are discharged, and the potential of the photoelectric conversion unit PD is reset to a certain value (VDD).

Next, the reset pulse ϕ PDRST is set to low, and the reset transistors for a photoelectric conversion unit Tr6 of all pixels are tuned off at the same time, and generation and accumulation of signal charges are started in the photoelectric conversion unit PD of all pixels. A signal charge is generated according to the light intensity of light incident on the photoelectric conversion unit PD, and the generated signal charge is accumulated in the well of the potential that can be generated by the effect of the pn junction in the photoelectric conversion unit PD. In this case, it is assumed that the signal charge stored in the charge accumulated capacitance part 153 is sequentially read during previous reading, and the charge accumulated capacitance part 153 becomes empty, but a timing at which the charge accumulated capacitance part 153 is reset may be provided separately.

Next, before a predetermined accumulation time has elapsed after the reset pulse ϕ PDRST is set to low, the first transfer pulse ϕ TRG1 is set to high, the first transfer transistors Tr1 of all pixels are turned on at the same time, and the signal charge stored in the photoelectric conversion unit PD is transferred to the impurity region 116. Then, since the impurity region 116, the impurity region 130, and the charge accumulated capacitance part 153 are electrically

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connected, the signal charge is temporarily accumulated in the impurity region **116**, the impurity region **130**, and the charge accumulated capacitance part **153** formed in the first substrate **180**. In addition, in this manner, while the first transfer pulse ϕ TRG1 is set to high, the signal charge is mainly accumulated in the charge accumulated capacitance part **153**.

Then, the first transfer pulse ϕ TRG1 is set to low and the first transfer transistors Tr1 of all pixels are turned off, and thus the signal charge mainly accumulated in the charge accumulated capacitance part **153** is transferred to the depletion layer of the impurity region **116** and the impurity region **130**. As shown in FIG. **26**, a time from when the reset pulse ϕ PDRST is set to low until the first transfer pulse ϕ TRG1 is set to low again is an accumulated exposure time (electronic shutter time). Here, when the first transfer pulse ϕ TRG1 is set to high, and the signal charge is transferred from the photoelectric conversion unit PD to the charge accumulated capacitance part **153**, the potential of the first transfer pulse ϕ TRG1 is set to a potential at which the signal charge from the photoelectric conversion unit PD can be completely transferred.

Next, the reset pulse ϕ PDRST is set to high, the reset transistors for a photoelectric conversion unit Tr6 of all pixels are turned on, and the photoelectric conversion unit PD is reset. Thereby, while the signal charge stored in the charge accumulated capacitance part **153** is read, the amount of the signal charge that is accumulated in the photoelectric conversion unit PD in excess of the maximum accumulated charge amount of the photoelectric conversion unit PD is prevented from overflowing to the charge accumulated capacitance part **153**. Alternatively, the photoelectric conversion unit PD is reset to the same potential as the power supply voltage VDD in preparation for the next signal charge accumulation. While the signal charge is accumulated in the charge accumulated capacitance part **153**, and the impurity regions **116** and **130**, as the potential of the first transfer pulse ϕ TRG1, a potential that forms an inversion layer on the front surface of the charge accumulated capacitance part **153** may be applied. Thereby, generation of a dark current can be minimized during accumulation of the signal charge.

Then, the selection pulse ϕ SEL(1) is set to high, the select transistor Tr1 in the first row is turned on, and the pixel **102** in the first row is selected. When the selection pulse ϕ SEL(1) in the first row is set to high, the reset pulse ϕ RST(1) is set to high, and the reset transistor Tr3 in the first row is turned on. Thereby, the potential of the impurity region **131**, which is a floating diffusion region connected to the gate electrode **140** of the amplifying transistor Tr4, is reset to the same potential as the power supply voltage VDD. In this case, the output when the amplifying transistor Tr4 is reset is stored in the column signal processing circuit **105** via the vertical signal line **109** (refer to FIG. **20**).

Next, the second transfer pulse ϕ TRG2(1) is set to high, the second transfer transistor Tr2 of the pixel **102** in the first row is turned on, and the signal charges in the impurity region **130** and the impurity region **116** of the pixel **102** in the first row are transferred to the impurity region **131**, which is a floating diffusion region. In this case, the potential of the second transfer pulse ϕ TRG2(1) is set to a potential at which the signal charge can be completely transferred from the impurity region **130** and the impurity region **116** to the impurity region **131**. When the signal charge is read in the impurity region **131**, the potential of the impurity region **131**, which is a floating diffusion region, changes, and the signal voltage according to the potential change is applied to

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the gate electrode **140** of the amplifying transistor Tr4. Therefore, the signal voltage amplified by the amplifying transistor Tr4 is output to the vertical signal line **109** (refer to FIG. **20**).

Therefore, the signal voltage output to the vertical signal line **109** (refer to FIG. **20**) is sent to the column signal processing circuit **105**. In the column signal processing circuit **105**, the difference between the output during reset stored earlier and the amplified signal voltage is output as a pixel signal of the pixel **102** in the first row. Therefore, when the horizontal transistor Tr7 is sequentially turned on by the horizontal drive circuit **106**, the pixel signals of the pixels **102** in the first row are serially output from the output terminal Vout through the output circuit **107** (refer to FIG. **20**).

Then, after the selection pulse ϕ SEL(1) is set to low, the selection pulse ϕ SEL(2) is set to high, the select transistor Tr5 in the second row is turned on, and the pixel **102** in the second row is selected. When the selection pulse ϕ SEL(2) of the select transistor Tr5 in the second row is set to high, the state of the second transfer pulse ϕ TRG2(2) and the reset pulse ϕ RST(2) is driven in the same manner as in the second transfer pulse ϕ TRG2(1) and the reset pulse ϕ RST(1) in the first row. Thereby, for the pixel **102** in the second row, the same read operation as in the first row described above is performed.

As can be understood from the above description, in the solid-state imaging device **101** of this modification example, the accumulated exposure time for generating and accumulating the signal charge in the photoelectric conversion unit PD is the same time for all pixels. That is, since each pixel **102** has the charge accumulated capacitance part **153**, the electronic shutter operation (global shutter operation) can be performed for all pixels at the same time. Therefore, the signal charges accumulated at the same time for all pixels are accumulated and kept in each charge accumulated capacitance part **153**, read in the impurity region **131** in line sequence, and the signal voltage amplified according to the potential of the signal charge is output via the vertical signal line **109** (refer to FIG. **20**).

An example of the solid-state imaging device to which the technology according to the present disclosure can be applied has been described above. The technology according to the present disclosure can be applied to the first substrate **180**, the second substrate **181**, the first connection electrode **127**, the first connection electrode **156** and the charge accumulated capacitance part **153** within the configuration described above. Specifically, the first substrate **2** in FIG. **1** can be applied to the first substrate **180** in FIG. **22**, the second substrate **3** in FIG. **1** can be applied to the second substrate **181**, the first electrode pad **8** in FIG. **1** can be applied to the first connection electrode **127** in FIG. **22**, the second electrode **17** in FIG. **1** can be applied to the first connection electrode **156** in FIG. **22**, and the second capacitor **20** in FIG. **1** can be applied to the charge accumulated capacitance part **153** in FIG. **22**. Thereby, according to the second capacitor **20** in FIG. **1**, the charge accumulated capacitance part **153** for accumulating the signal charge generated in the photoelectric conversion unit PD when the global shutter operation is executed can be formed. When the technology according to the present disclosure is applied to the first substrate **180**, the second substrate **181**, the first connection electrode **127**, the first connection electrode **156** and the charge accumulated capacitance part **153**, since deterioration of performance of the capacitor (the charge

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accumulated capacitance part **153**) due to the bonding process can be minimized, a better captured image can be obtained.

Here, in this modification example, an example in which the charge accumulated capacitance part **153** is formed in the region for substrate connection **195** of the second substrate **181** is shown, but other configurations can be used. For example, it may be formed in the region for substrate connection **190** of the first substrate **180**. When formed in the region for substrate connection **190**, the first capacitor **10** in FIG. **1** can be applied to the charge accumulated capacitance part **153**. Thereby, according to the first capacitor **10** in FIG. **1**, the charge accumulated capacitance part **153** for accumulating the signal charge generated in the photoelectric conversion unit PD when the global shutter operation is executed can be formed.

3. Example of Application to Electronic Device

The technology according to the present disclosure (the present technology) may be applied to various electronic devices, for example, an imaging device such as a digital still camera and a digital video camera, a mobile phone having an imaging function, or another device having an imaging function.

FIG. **27** is a diagram showing an example of a schematic configuration of an imaging device as an electronic device to which the technology according to the present disclosure (the present technology) can be applied.

As shown in FIG. **27**, an imaging device **201** includes an optical system **202**, a shutter device **203**, a solid-state imaging element **204**, a drive circuit **205**, a signal processing circuit **206**, a monitor **207**, and a memory **208**, and can capture still images and moving images.

The optical system **202** includes one or more lenses, and guides light (incident light) from a subject to the solid-state imaging element **204**, and forms an image on the light receiving surface of the solid-state imaging element **204**.

The shutter device **203** is arranged between the optical system **202** and the solid-state imaging element **204**, and controls a light emission period and a light-blocking period for the solid-state imaging element **204** according to the control of the drive circuit **205**.

The solid-state imaging element **204** is formed by a package including the above solid-state imaging element **204**. The solid-state imaging element **204** accumulates signal charges for a certain period according to light formed into an image on the light receiving surface via the optical system **202** and the shutter device **203**. The signal charge accumulated in the solid-state imaging element **204** is transferred according to the drive signal (timing signal) supplied from the drive circuit **205**. The drive circuit **205** outputs a drive signal that controls the transfer operation of the solid-state imaging element **204** and the shutter operation of the shutter device **203**, and drives the solid-state imaging element **204** and the shutter device **203**. The signal processing circuit **206** performs various types of signal processing on the signal charge output from the solid-state imaging element **204**. The image (image data) obtained by signal processing performed by the signal processing circuit **206** is supplied to and displayed on the monitor **207**, and supplied and stored (recorded) in the memory **208**.

An example of the electronic device (imaging device) to which the technology according to the present disclosure can be applied has been described above. The technology according to the present disclosure can be applied to the solid-state imaging element **204** within the configuration

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described above. Specifically, the image sensor in FIG. **21** and FIG. **22** can be applied to the solid-state imaging element **204**. When the technology according to the present disclosure is applied to the solid-state imaging element **204**, since deterioration of performance of the capacitor due to the bonding process can be minimized, a better captured image can be obtained.

4. Example of Application to Moving Object

The technology according to the present disclosure (the present technology) may be applied to devices mounted in any type of moving objects, for example, automobiles, electric vehicles, hybrid electric vehicles, motorcycles, bicycles, personal mobilities, airplanes, drones, ships, and robots.

FIG. **28** is a block diagram showing a schematic configuration example of a vehicle control system, which is an example of a moving object control system to which the technology according to the present disclosure can be applied.

A vehicle control system **12000** includes a plurality of electronic control units connected via a communication network **12001**. In the example shown in FIG. **28**, the vehicle control system **12000** includes a drive system control unit **12010**, a body system control unit **12020**, an outside-vehicle information detection unit **12030**, an in-vehicle information detection unit **12040**, and an integrated control unit **12050**. In addition, as the functional configuration of the integrated control unit **12050**, a microcomputer **12051**, an audio image output unit **12052**, and an in-vehicle network interface (I/F) **12053** are shown.

The drive system control unit **12010** controls the operation of a device related to a vehicle drive system according to various programs. For example, the drive system control unit **12010** functions as a control device for a driving force generating device for generating a driving force of a vehicle such as an internal combustion engine or a drive motor, a driving force transmission mechanism for transmitting a driving force to wheels, a steering mechanism for adjusting a steering angle of a vehicle, and a braking device for generating a braking force of a vehicle.

The body system control unit **12020** controls operations of various devices mounted in a vehicle body according to various programs. For example, the body system control unit **12020** functions as a control device such as a keyless entry system, a smart key system, a power window device, or various lamps such as a headlamp, a back lamp, a brake lamp, a turn signal and a fog lamp. In this case, radio waves transmitted from a portable device that substitutes for a key or signals of various switches may be input to the body system control unit **12020**. The body system control unit **12020** receives such a radio wave or signal input, and controls a door lock device, a power window device, and a lamp of the vehicle.

The outside-vehicle information detection unit **12030** detects information outside the vehicle in which the vehicle control system **12000** is mounted. For example, an imaging unit **12031** is connected to the outside-vehicle information detection unit **12030**. The outside-vehicle information detection unit **12030** causes the imaging unit **12031** to capture an image of the outside of the vehicle and receives the captured image. The outside-vehicle information detection unit **12030** may perform object detection processing or distance detection processing for peoples, cars, obstacles, signs, and letters on the road based on the received image.

The imaging unit **12031** is an optical sensor that receives light and outputs an electrical signal according to the intensity of the light received. The imaging unit **12031** can output an electrical signal as an image or output it as a distance measurement information. In addition, the light received by the imaging unit **12031** may be visible light or invisible light such as infrared rays.

The in-vehicle information detection unit **12040** detects information in the vehicle. In the in-vehicle information detection unit **12040**, for example, a driver status detection unit **12041** that detects the driver's status is connected. The driver status detection unit **12041** includes, for example, a camera that images the driver, and the in-vehicle information detection unit **12040** may calculate the degree of fatigue or degree of concentration of the driver based on detection information input from the driver status detection unit **12041**, and may determine whether the driver is asleep.

The microcomputer **12051** can calculate a control target value of the driving force generating device, the steering mechanism or the braking device based information inside and outside the vehicle acquired by the outside-vehicle information detection unit **12030** or the in-vehicle information detection unit **12040**, and output a control command to the drive system control unit **12010**. For example, the microcomputer **12051** can perform cooperative control in order to realize functions of an advanced driver assistance system (ADAS) such as following traveling, vehicle speed maintenance driving, vehicle collision warning, and vehicle lane deviation warning based on vehicle collision avoidance, impact mitigation, and inter-vehicle distance.

In addition, the microcomputer **12051** can perform cooperative control for automatic driving in which autonomous driving is performed without the operation of the driver by controlling the driving force generating device, the steering mechanism or the braking device based on information around the vehicle acquired by the outside-vehicle information detection unit **12030** or the in-vehicle information detection unit **12040**.

In addition, the microcomputer **12051** can output a control command to the body system control unit **12020** based on the information outside the vehicle acquired by the outside-vehicle information detection unit **12030**. For example, the microcomputer **12051** can perform cooperative control for antilglare such as switching a high beam to a low beam by controlling a headlamp according to a position of a preceding vehicle or an oncoming vehicle detected by the outside-vehicle information detection unit **12030**.

The audio image output unit **12052** transmits an output signal of at least one of audio and an image to an output device that can visually or audibly notify the passenger of the vehicle or the outside of information. In the example shown in FIG. 28, as such an output device, an audio speaker **12061**, a display unit **12062** and an instrument panel **12063** are shown. The display unit **12062** may include, for example, at least one of an onboard display and a head-up display.

FIG. 29 is a diagram showing an example of an installation position of the imaging unit **12031**.

In FIG. 29, a vehicle **12100** includes imaging units **12101**, **12102**, **12103**, **12104**, and **12105** as the imaging unit **12031**.

The imaging units **12101**, **12102**, **12103**, **12104**, and **12105** are provided at positions, for example, the front nose, side mirrors, rear bumpers, back doors and the upper part of the windshield in the cabin of the vehicle **12100**. The imaging unit **12101** provided in the front nose and the imaging unit **12105** provided in the upper part of the windshield in the cabin mainly acquire an image to the front

of the vehicle **12100**. The imaging units **12102** and **12103** provided in the side mirrors mainly acquire an image to the side of the vehicle **12100**. The imaging unit **12104** provided in the rear bumper or the back door mainly acquires an image to the rear of the vehicle **12100**. The front images acquired by the imaging units **12101** and **12105** are mainly used for detecting preceding vehicles, pedestrians, obstacles, traffic lights, traffic signs, lanes and the like.

Here, FIG. 29 shows an example of imaging ranges of the imaging units **12101** to **12104**. An imaging range **12111** indicates an imaging range of the imaging unit **12101** provided in the front nose, imaging ranges **12112** and **12113** indicate imaging ranges of the imaging units **12102** and **12103** provided in the side mirror, respectively, and an imaging range **12114** indicates an imaging range of the imaging unit **12104** provided in the rear bumper or back door. For example, a bird's-eye view image of the vehicle **12100** from above can be obtained by superimposing image data captured by the imaging units **12101** to **12104**.

At least one of the imaging units **12101** to **12104** may have a function of acquiring distance information. For example, at least one of the imaging units **12101** to **12104** may be a stereo camera including a plurality of imaging elements or an imaging element having pixels for phase difference detection.

For example, the microcomputer **12051** determines a distance to each three-dimensional object in the imaging ranges **12111** to **12114**, and a change in the distance over time (a relative speed with respect to the vehicle **12100**) based on the distance information obtained from the imaging units **12101** to **12104**, and particularly, can extract a three-dimensional object that travels at a predetermined speed (for example, 0 km/h or more) in almost the same direction as the vehicle **12100**, which is the closest three-dimensional object on the traveling path of the vehicle **12100**, as a preceding vehicle. In addition, the microcomputer **12051** can set an inter-vehicle distance to be secured in advance in front of the preceding vehicle, and perform automatic brake control (including following stop control), automatic acceleration control (including following start control), and the like. In this manner, it is possible to perform cooperative control for automatic driving in which autonomous driving is performed without the operation of the driver.

For example, based on the distance information obtained from the imaging units **12101** to **12104**, the microcomputer **12051** classifies the three-dimensional object data related to the three-dimensional object as a two-wheeled vehicle, an ordinary vehicle, a large vehicle, a pedestrian, a utility pole or another three-dimensional object and performs extraction, and can use the result for automatic avoidance of an obstacle. For example, the microcomputer **12051** distinguishes obstacle around the vehicle **12100** as obstacles that the driver of the vehicle **12100** can visually recognize and obstacles that are difficult for the driver to visually recognize. Therefore, the microcomputer **12051** determines collision risk indicating the degree of risk of collision with each obstacle, and when the collision risk is equal to or higher than the set value and there is a possibility of collision, an alarm is output to the driver through the audio speaker **12061** and the display unit **12062**, forced deceleration and avoidance steering are performed through the drive system control unit **12010**, and thus it is possible to perform driving support for collision avoidance.

At least one of the imaging units **12101** to **12104** may be an infrared camera that detects infrared rays. For example, the microcomputer **12051** can recognize a pedestrian by determining whether there is a pedestrian in the captured

image of the imaging units **12101** to **12104**. Such pedestrian recognition is performed by, for example, a procedure in which feature points in the captured images of the imaging units **12101** to **12104** as infrared cameras are extracted and a procedure in which pattern matching processing is performed on a series of feature points indicating the outline of the object and it is determined whether the object is a pedestrian. When the microcomputer **12051** determines that there is a pedestrian in the captured images of the imaging units **12101** to **12104**, and the pedestrian is recognized, the audio image output unit **12052** controls the display unit **12062** so that the recognized pedestrian is superimposed and displayed with a square contour line for emphasis. In addition, the audio image output unit **12052** may control the display unit **12062** so that an icon indicating a pedestrian or the like is displayed at a desired position.

An example of the vehicle control system to which the technology according to the present disclosure can be applied has been described above. The technology according to the present disclosure can be applied to the imaging unit **12031** within the configuration described above. Specifically, the image sensor in FIG. **21** and FIG. **22** can be applied to the imaging unit **12031**. When the technology according to the present disclosure is applied to the imaging unit **12031**, since deterioration of the performance of the capacitor due to the bonding process can be minimized, a better captured image can be obtained, and fatigue of the driver can be reduced.

5. Example of Application to Endoscopic Surgery System

The technology according to the present disclosure (the present technology) may be applied to, for example, an endoscopic surgery system.

FIG. **30** is a diagram showing an example of a schematic configuration of an endoscopic surgery system to which the technology according to the present disclosure (the present technology) can be applied.

FIG. **30** shows a case in which a surgeon (doctor) **11131** performs surgery on a patient **11132** on a patient bed **11133** using an endoscopic surgery system **11000**. As shown, the endoscopic surgery system **11000** is composed of an endoscope **11100**, another surgical tool **11110** such as a pneumoperitoneum tube **11111** and an energy treatment tool **11112**, a support arm device **11120** that supports the endoscope **11100**, and a cart **11200** in which various devices for endoscopic surgery are mounted.

The endoscope **11100** is composed of a lens barrel **11101** in which a region having a predetermined length from the tip is inserted into the body cavity of the patient **11132** and a camera head **11102** connected to the base end of the lens barrel **11101**. In the shown example, the endoscope **11100** configured as a so-called rigid endoscope having the rigid lens barrel **11101** is shown, but the endoscope **11100** may be configured as a so-called soft endoscope having a soft lens barrel.

An opening into which an objective lens is fitted is provided at the tip of the lens barrel **11101**. A light source device **11203** is connected to the endoscope **11100**, and light generated by the light source device **11203** is guided to the tip of the lens barrel by the light guide extending inside the lens barrel **11101**, and emitted to an observation target in the body cavity of the patient **11132** via the objective lens. Here, the endoscope **11100** may be a direct endoscope or may be a perspective endoscope or a side endoscope.

An optical system and an imaging element are provided inside the camera head **11102**, and reflected light (observation light) from the observation target is condensed on the imaging element by the optical system. The observation light is photoelectrically converted by the imaging element, and an electrical signal corresponding to the observation light, that is, an image signal corresponding to the observation image is generated. The image signal is transmitted as RAW data to a camera control unit (CCU) **11201**.

The CCU **11201** is composed of a central processing unit (CPU), a graphics processing unit (GPU) or the like, and comprehensively controls the operation of the endoscope **11100** and a display device **11202**. In addition, the CCU **11201** receives an image signal from the camera head **11102**, and performs various types of image processing for displaying an image based on the image signal, for example, development processing (demosaic processing) on the image signal.

The display device **11202** displays an image based on the image signal on which image processing is performed by the CCU **11201** according to the control from the CCU **11201**.

The light source device **11203** is composed of, for example, a light source such as a light emitting diode (LED), and supplies irradiation light when a surgical part or the like is imaged to the endoscope **11100**.

An input device **11204** is an input interface for the endoscopic surgery system **11000**. A user can input various types of information and input an instruction to the endoscopic surgery system **11000** via the input device **11204**. For example, the user inputs an instruction to change imaging conditions (the type of irradiation light, a magnification, a focal distance and the like) and the like by the endoscope **11100**.

A treatment tool control device **11205** controls driving of the energy treatment tool **11112** for tissue cauterization, incision or blood vessel sealing. In order to secure a field of view of the endoscope **11100** and secure an operation space of the surgeon, a pneumoperitoneum device **11206** sends a gas into a body cavity via the pneumoperitoneum tube **11111** in order to inflate the body cavity of the patient **11132**. A recorder **11207** is a device that can record various types of information related to surgery. A printer **11208** is a device that can print various types of information related to surgery in various formats such as text, images and graphs.

Here, the light source device **11203** that supplies irradiation light when a surgical part is imaged to the endoscope **11100** can be composed of, for example, an LED, a laser light source or a white light source composed of a combination thereof. When a white light source is composed of a combination of RGB laser light sources, since the output intensity of each color (each wavelength) and the output timing can be controlled with high accuracy, white balance of the captured image can be adjusted in the light source device **11203**. In addition, in this case, laser light from each of the RGB laser light sources is emitted to the observation target in a time-division manner, driving of the imaging element of the camera head **11102** is controlled in synchronization with the emission timing, and thus an image corresponding to each RGB can be captured in a time-division manner. According to the method, a color image can be obtained without providing a color filter in the imaging element.

In addition, driving of the light source device **11203** may be controlled so that the intensity of the output light changes at predetermined times. When driving of the imaging element of the camera head **11102** is controlled in synchronization with the timing of the change in the light intensity, an

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image is acquired in a time-division manner, and the image is synthesized, a so-called image in a high dynamic range without underexposure or overexposure can be generated.

In addition, the light source device **11203** may have a configuration in which light in a predetermined wavelength band corresponding to special light observation can be supplied. In the special light observation, for example, by emitting light in a narrower band than irradiation light (that is, white light) during normal observation using wavelength dependence of light absorption in a body tissue, so-called narrow band light observation (narrow band imaging) in which a predetermined tissue such as a blood vessel in the mucous membrane surface layer is imaged with a high contrast is performed. Alternatively, in the special light observation, fluorescence observation in which an image is obtained by fluorescence generated by emitting excitation light may be performed. The fluorescence observation can be performed by emitting excitation light to a body tissue, and observing fluorescence from the body tissue (autofluorescence observation), or locally injecting a reagent such as indocyanine green (ICG) to a body tissue, and emitting excitation light corresponding to a fluorescence wavelength of the reagent to the body tissue to obtain a fluorescence image. The light source device **11203** can supply narrow band light and/or excitation light corresponding to such special light observation.

FIG. **31** is a block diagram showing an example of a functional configuration of the camera head **11102** and the CCU **11201** shown in FIG. **30**.

The camera head **11102** includes a lens unit **11401**, an imaging unit **11402**, a drive unit **11403**, a communication unit **11404**, and a camera head control unit **11405**. The CCU **11201** includes a communication unit **11411**, an image processing unit **11412**, and a control unit **11413**. The camera head **11102** and the CCU **11201** are connected to each other via a transmission cable **11400** so that they can communicate with each other.

The lens unit **11401** is an optical system provided at the connection part with respect to the lens barrel **11101**. Observation light taken from the tip of the lens barrel **11101** is guided to the camera head **11102** and enters the lens unit **11401**. The lens unit **11401** is composed of a combination of a plurality of lenses including a zoom lens and a focus lens.

The imaging unit **11402** is composed of an imaging element. The imaging element constituting the imaging unit **11402** may be one element (so-called single plate type) or a plurality of elements (so-called multi-plate type). When the imaging unit **11402** is composed of a multi-plate type, for example, image signals corresponding to RGBs are generated by the imaging elements, and synthesized, and thereby a color image may be obtained. Alternatively, the imaging unit **11402** may include a pair of imaging elements for acquiring image signals for the right eye and the left eye corresponding to 3D (dimensional) display. When 3D display is performed, the surgeon **11131** can determine the depth of biological tissues in the surgical part more accurately. Here, when the imaging unit **11402** is composed of a multi-plate type, a plurality of lens units **11401** may be provided according to the imaging elements.

In addition, the imaging unit **11402** need not necessarily be provided in the camera head **11102**. For example, the imaging unit **11402** may be provided inside the lens barrel **11101**, immediately after the objective lens.

The drive unit **11403** is composed of an actuator, and moves the zoom lens and the focus lens of the lens unit **11401** along the optical axis by a predetermined distance according to the control from the camera head control unit

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11405. Thereby, the magnification and the focus of the image captured by the imaging unit **11402** can be appropriately adjusted.

The communication unit **11404** is composed of a communication device for transmitting and receiving various types of information to and from the CCU **11201**. The communication unit **11404** transmits the image signal obtained from the imaging unit **11402** as RAW data to the CCU **11201** via the transmission cable **11400**.

In addition, the communication unit **11404** receives a control signal for controlling driving of the camera head **11102** from the CCU **11201** and supplies it to the camera head control unit **11405**. The control signal includes information related to imaging conditions, for example, information specifying a frame rate of a captured image, information specifying an exposure value during imaging, and/or information specifying a magnification and a focus of a captured image.

Here, imaging conditions such as a frame rate, an exposure value, a magnification, and a focus may be appropriately specified by the user, or may be automatically set by the control unit **11413** of the CCU **11201** based on the acquired image signal. In the latter case, a so-called auto exposure (AE) function, auto focus (AF) function and auto white balance (AWB) function are provided in the endoscope **11100**.

The camera head control unit **11405** controls driving of the camera head **11102** based on the control signal received from the CCU **11201** via the communication unit **11404**.

The communication unit **11411** is composed of a communication device for transmitting and receiving various types of information to and from the camera head **11102**. The communication unit **11411** receives the image signal transmitted from the camera head **11102** via the transmission cable **11400**.

In addition, the communication unit **11411** transmits a control signal for controlling driving of the camera head **11102** to the camera head **11102**. The image signal and the control signal can be transmitted through telecommunication, optical communication or the like.

The image processing unit **11412** performs various types of image processing on the image signal, which is RAW data transmitted from the camera head **11102**.

The control unit **11413** performs various controls related to imaging of a surgical part and the like with the endoscope **11100** and displaying of the captured image obtained by imaging a surgical part and the like. For example, the control unit **11413** generates a control signal for controlling driving of the camera head **11102**.

In addition, the control unit **11413** causes the display device **11202** to display a captured image showing the surgical part and the like based on the image signal on which image processing is performed by the image processing unit **11412**. In this case, the control unit **11413** may recognize various objects in the captured image using various image recognition technologies. For example, the control unit **11413** can recognize surgical tools such as forceps, specific biological parts, bleeding, mist when the energy treatment tool **11112** is used and the like by detecting the edge shape and color of the object included in the captured image. When the control unit **11413** causes the display device **11202** to display the captured image, it may cause various types of surgical support information to be superimposed and displayed with the image of the surgical part using the recognition result. When the surgical support information is superimposed and displayed, and presented to the surgeon

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11131, it is possible to reduce the burden on the surgeon 11131 and the surgeon 11131 can reliably proceed the surgery.

The transmission cable 11400 connecting the camera head 11102 and the CCU 11201 is an electrical signal cable supporting electrical signal communication, an optical fiber supporting optical communication, or a composite cable thereof.

Here, in the shown example, communication is performed using the transmission cable 11400 in a wired manner, but communication between the camera head 11102 and the CCU 11201 may be performed in a wireless manner.

An example of the endoscopic surgery system to which the technology according to the present disclosure can be applied has been described above. The technology according to the present disclosure can be applied to the imaging unit 11402 within the configuration described above. Specifically, the image sensor in FIG. 21 and FIG. 22 can be applied to the imaging unit 11402. When the technology according to the present disclosure is applied to the imaging unit 11402, since deterioration of the performance of the capacitor due to the bonding process can be minimized, a clearer surgical part image can be obtained, and the surgeon can reliably confirm the surgical part.

While the endoscopic surgery system has been described here as an example, the technology according to the present disclosure may be applied to other systems, for example, a microscopic surgery system.

In addition, the present technology may have the following configuration.

- (1) A semiconductor device, including
 - a first substrate;
 - a second substrate bonded to the first substrate;
 - a first electrode which is provided in the first substrate and of which one surface is positioned on the same surface as a bonding surface between the first substrate and the second substrate;
 - a second electrode which is provided in the second substrate and of which one surface is positioned on the same surface as the bonding surface and bonded to one surface of the first electrode; and
 - at least one of a first capacitor which is provided in the first substrate and of which one electrode is electrically connected to the other surface of the first electrode and a second capacitor which is provided in the second substrate and of which one electrode is electrically connected to the other surface of the second electrode.
- (2) The semiconductor device according to (1), wherein the first electrode and the second electrode are copper electrodes made of pure copper or copper alloy.
- (3) The semiconductor device according to (1) or (2), wherein the first electrode includes a first electrode pad of which one surface is positioned on the same surface as the bonding surface and a first via extending from the other surface of the first electrode pad, and wherein the second electrode includes a second electrode pad of which one surface is positioned on the same surface as the bonding surface and a second via extending from the other surface of the second electrode pad.
- (4) The semiconductor device according to any one of (1) to (3), wherein a plurality of each of the first electrodes and the second electrodes are provided,

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wherein a plurality of at least one of first capacitors and second capacitors are provided, and wherein two or more of the plurality of first capacitors or two or more of the plurality of second capacitors are electrically connected to the same wiring.

- (5) The semiconductor device according to any one of (1) to (4), wherein at least one of the first capacitor and the second capacitor includes two electrodes that face each other and an insulating film arranged between the two electrodes.
- (6) The semiconductor device according to any one of (1) to (4), wherein at least one of the first capacitor and the second capacitor forms a multilayer structure in which the electrode and the insulating film are alternately and repeatedly arranged.
- (7) The semiconductor device according to any one of (1) to (4), wherein at least one of the first capacitor and the second capacitor includes a bottom part and a side wall part extending from an edge part of the bottom part in a direction intersecting the bottom part.
- (8) The semiconductor device according to any one of (1) to (7), wherein the material of the electrode of the first capacitor and the second capacitor is tantalum, tantalum nitride, titanium, titanium nitride, tungsten nitride, zirconium nitride or cobalt.
- (9) The semiconductor device according to any one of (1) to (8), wherein an insulating film of the first capacitor and the second capacitor is a single-layer film composed of any of tantalum oxide, hafnium oxide, aluminum oxide, silicon nitride and zirconium oxide, or a multilayer film formed by combining these.
- (10) A solid-state imaging device, including:
 - a sensor substrate in which a plurality of photoelectric conversion units are arranged;
 - a logic substrate which is bonded to the sensor substrate and in which a circuit that processes an electrical signal from the photoelectric conversion unit is integrated;
 - a first electrode which is provided in the sensor substrate and of which one surface is positioned on the same surface as a bonding surface between the sensor substrate and the logic substrate;
 - a second electrode which is provided in the logic substrate and of which one surface is positioned on the same surface as the bonding surface and bonded to one surface of the first electrode; and
 - at least one of a first capacitor which is provided in the sensor substrate and of which one electrode is electrically connected to the other surface of the first electrode and a second capacitor which is provided in the logic substrate and of which one electrode is electrically connected to the other surface of the second electrode.
- (11) The solid-state imaging device according to (10), wherein at least one of the first capacitor and the second capacitor constitutes a charge accumulated capacitance

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part for accumulating a signal charge generated by the photoelectric conversion unit when a global shutter operation is executed.

(12)

An electronic device, including
a solid-state imaging device including a sensor substrate in which a plurality of photoelectric conversion units are arranged, a logic substrate which is bonded to the sensor substrate and in which a circuit that processes an electrical signal from the photoelectric conversion unit is integrated, a first electrode which is provided in the sensor substrate and of which one surface is positioned on the same surface as a bonding surface between the sensor substrate and the logic substrate, a second electrode which is provided in the logic substrate and of which one surface is positioned on the same surface as the bonding surface and bonded to one surface of the first electrode, and at least one of a first capacitor which is provided in the sensor substrate and of which one electrode is electrically connected to the other surface of the first electrode and a second capacitor which is provided in the logic substrate and of which one electrode is electrically connected to the other surface of the second electrode;
an optical lens that forms image light from a subject into an image on an imaging surface of the solid-state imaging device; and
a signal processing circuit that performs signal processing on a signal output from the solid-state imaging device.

REFERENCE SIGNS LIST

1 Semiconductor device
2 First substrate
3 Second substrate
4 Interlayer insulating film
5a First wiring layer
5b Second wiring layer
5c Third wiring layer
5d Fourth wiring layer
6 Interlayer insulating film
7, 7a, 7b First electrode
8 First electrode pad
9 First via
10 First capacitor
11 Electrode
12 Electrode
13 Insulating film
14 Interlayer insulating film
15a First wiring layer
15b Second wiring layer
15c Third wiring layer
15d Fourth wiring layer
15e Fifth wiring layer
15f Sixth wiring layer
16 Interlayer insulating film
17, 17a, 17b Second electrode
18 Second electrode pad
19 Second via
20 Second capacitor
21 Electrode
22 Electrode
23 Insulating film
24 Region for substrate connection
25 Pixel region
26 I/O unit
27 Driver

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28, 29 Insulating film
30, 30a to 30e Electrode
31, 32 Insulating film
33 Bottom part
34, 35 Side wall part
36 Color filter
37 Micro lens

The invention claimed is:

1. A semiconductor device, comprising:

a first substrate that includes a top surface and a bottom surface;
a second substrate that includes a top surface and a bottom surface, wherein the top surface of the second substrate is bonded to the bottom surface of the first substrate;
a first electrode in the first substrate, wherein the first electrode includes a top surface and a bottom surface, and the bottom surface of the first electrode is on the bottom surface of the first substrate;
a second electrode in the second substrate, wherein the second electrode includes a top surface and a bottom surface, the top surface of the second electrode is in contact with the top surface of the second substrate, and the second electrode is bonded to the bottom surface of the first electrode;
a first capacitor in contact with the top surface of the first substrate, wherein the first capacitor includes an electrode that is electrically connected to the top surface of the first electrode; and
a second capacitor in contact with the bottom surface of the second substrate, wherein the second capacitor includes an electrode that is electrically connected to the top surface of the second electrode.

2. The semiconductor device according to claim 1, wherein the first electrode and the second electrode are copper electrodes that include one of pure copper or copper alloy.

3. The semiconductor device according to claim 1, wherein

the first electrode further includes a first electrode pad that includes a top surface and a bottom surface, the bottom surface of the first electrode pad is on the bottom surface of the first substrate, a first via extends from the top surface of the first electrode pad, the second electrode further includes a second electrode pad that includes a top surface and a bottom surface, the top surface of the second electrode pad is in contact with the top surface of the second substrate, and a second via extends from the bottom surface of the second electrode pad.

4. The semiconductor device according to claim 1, further comprising:

a plurality of first electrodes and a plurality of second electrodes, wherein the plurality of first electrodes includes the first electrode, and the plurality of second electrodes includes the second electrode; and
a plurality of first capacitors and a plurality of second capacitors, wherein the plurality of first capacitors includes the first capacitor, the plurality of second capacitors includes the second capacitor, and

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one of at least two of the plurality of first capacitors or at least two of the plurality of second capacitors are electrically connected to a same wiring.

5. The semiconductor device according to claim 1, wherein at least one of the first capacitor or the second capacitor further includes two electrodes with an insulating film between the two electrodes.

6. The semiconductor device according to claim 1, wherein at least one of the first capacitor or the second capacitor further includes a multilayer structure in which an electrode and an insulating film are alternately and repeatedly arranged.

7. The semiconductor device according to claim 1, wherein at least one of the first capacitor or the second capacitor further includes a bottom part and a side wall part that extends from an edge part of the bottom part in a direction that intersects the bottom part.

8. The semiconductor device according to claim 1, wherein a material of each of the electrode of the first capacitor and the electrode of the second capacitor is one of tantalum, tantalum nitride, titanium, titanium nitride, tungsten nitride, zirconium nitride, or cobalt.

9. The semiconductor device according to claim 1, wherein an insulating film of the first capacitor and the second capacitor includes one of

a single-layer film that includes at least one of tantalum oxide, hafnium oxide, aluminum oxide, silicon nitride, or zirconium oxide, or

a multilayer film that includes a combination of two or more materials selected from the group consisting of tantalum oxide, hafnium oxide, aluminum oxide, silicon nitride, and zirconium oxide.

10. A solid-state imaging device, comprising:

a sensor substrate which includes a plurality of photoelectric conversion units, a top surface, and a bottom surface;

a logic substrate that includes a top surface and a bottom surface, wherein

the top surface of the logic substrate is bonded to the bottom surface of the sensor substrate, and the logic substrate further includes a circuit configured to process electrical signals from the plurality of photoelectric conversion units;

a first electrode in the sensor substrate, wherein the first electrode includes a top surface and a bottom surface, and

the bottom surface of the first electrode is on the bottom surface of the sensor substrate;

a second electrode in the logic substrate, wherein the second electrode includes a top surface and a bottom surface,

the top surface of the second electrode is in contact with the top surface of the logic substrate, and

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the second electrode is bonded to the bottom surface of the first electrode; and

a first capacitor in contact with the top surface of the sensor substrate, wherein the first capacitor includes an electrode that is electrically connected to the top surface of the first electrode; and

a second capacitor in contact with the bottom surface of the logic substrate, wherein the second capacitor includes an electrode that is electrically connected to the top surface of the second electrode.

11. The solid-state imaging device according to claim 10, wherein at least one of the first capacitor or the second capacitor constitutes a charge accumulated capacitance part that accumulates signal charge generated by the plurality of photoelectric conversion units in case of execution of a global shutter operation.

12. An electronic device, comprising:

a solid-state imaging device including

a sensor substrate which includes a plurality of photoelectric conversion units a top surface, and a bottom surface,

a logic substrate that includes a top surface and a bottom surface, wherein

the top surface of the logic substrate is bonded to the bottom surface of the sensor substrate, and the logic substrate further includes a circuit configured to process electrical signals from the plurality of photoelectric conversion units,

a first electrode in the sensor substrate, wherein the first electrode includes a top surface and a bottom surface, and

the bottom surface of the first electrode is on the bottom surface of the sensor substrate,

a second electrode in the logic substrate, wherein the second electrode includes a top surface and a bottom surface, and

the top surface of the second electrode is in contact with the top surface of the logic substrate,

a first capacitor in contact with the top surface of the sensor substrate, wherein the first capacitor includes an electrode that is electrically connected to the top surface of the first electrode, and

a second capacitor in contact with the bottom surface of the logic substrate, wherein the second capacitor includes an electrode that is electrically connected to the top surface of the second electrode;

an optical lens configured to form image light from a subject into an image on an imaging surface of the solid-state imaging device; and

a signal processing circuit configured to execute a signal processing operation on a signal output from the solid-state imaging device.

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